Vision-Language-Action Models for Autonomous Driving: Past, Present, and Future

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Autonomous driving has long relied on modular "Perception-Decision-Action" pipelines, where hand-crafted interfaces and rule-based components often break down in complex or long-tailed scenarios. Their cascaded design further propagates perception errors, degrading downstream planning and control. **Vision-Action (VA)** models address some limitations by learning direct mappings from visual inputs to actions, but they remain opaque, sensitive to distribution shifts, and lack structured reasoning or instruction-following capabilities. Recent progress in Large Language Models (LLMs) and multimodal learning has motivated the emergence of **Vision-Language-Action (VLA)** frameworks, which integrate perception with language-grounded decision making. By unifying visual understanding, linguistic reasoning, and actionable outputs, VLAs offer a more interpretable, generalizable, and human-aligned paradigm for driving policies. This work provides a structured characterization of the emerging VLA landscape for autonomous driving. We trace the evolution from early VA approaches to modern VLA frameworks and organize existing methods into two principal paradigms: *End-to-End VLA*, which integrates perception, reasoning, and planning within a single model, and *Dual-System VLA*, which separates slow deliberation (via VLMs) from fast, safety-critical execution (via planners). Within these paradigms, we further distinguish subclasses such as textual *vs.* numerical action generators and explicit *vs.* implicit guidance mechanisms. We also summarize representative datasets and benchmarks for evaluating VLA-based driving systems and highlight key challenges and open directions, including robustness, interpretability, and instruction fidelity. Overall, this work aims to establish a coherent foundation for advancing human-compatible autonomous driving systems.

Project Page: https://worldbench.github.io/vla4ad

GitHub Repo: https://github.com/worldbench/awesome-vla-for-ad

HuggingFace Leaderboard: https://huggingface.co/spaces/worldbench/vla4ad

1 Introduction

The pursuit of fully autonomous driving (AD) has long been a central goal in AI and robotics [1–3]. Conventional AD systems typically adopt a modular "Perception-Decision-Action" pipeline, where mapping [4, 5], object detection [6–9], motion prediction [10–12], and trajectory planning [13, 14] are developed and optimized as separate components. While this design has achieved strong performance in structured environments, its reliance on hand-crafted interfaces and rules limits adaptability in complex [15–17], dynamic [18–20], and

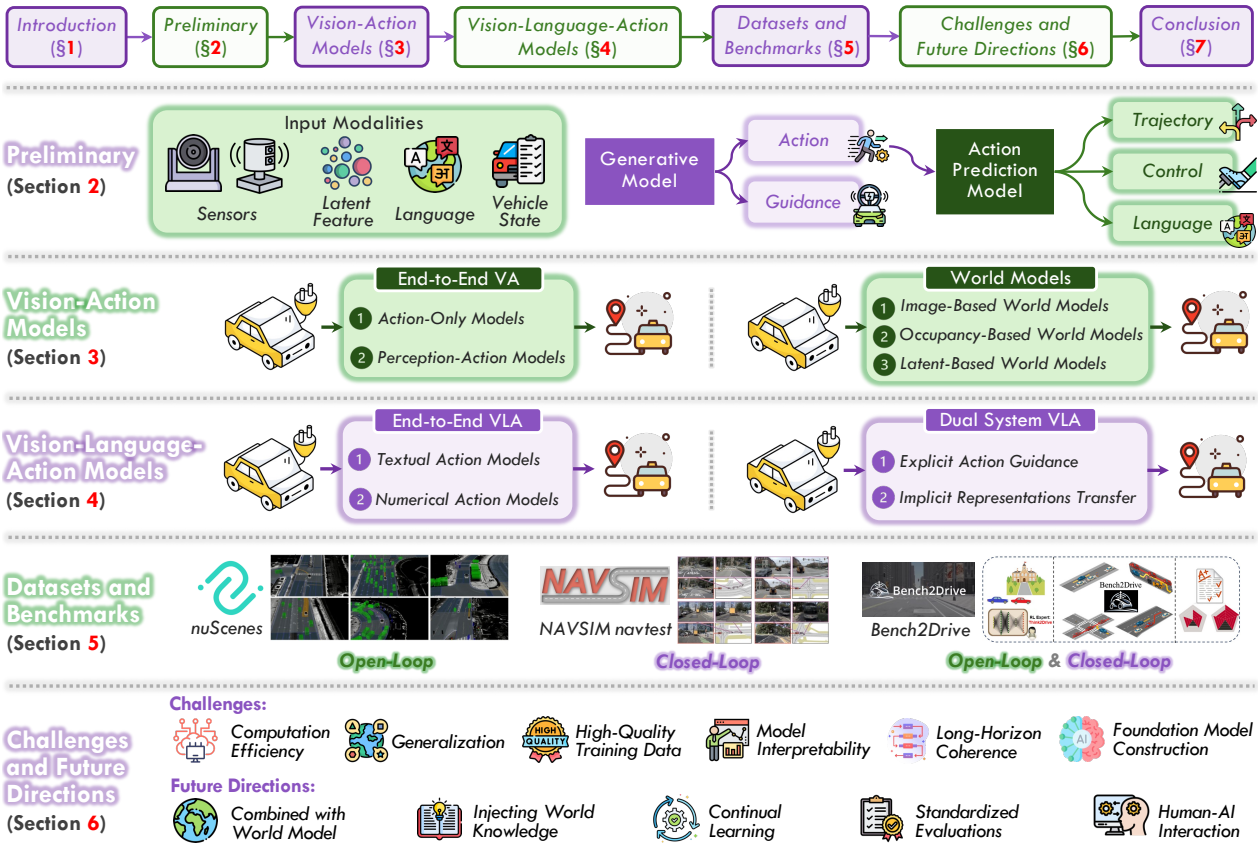


Figure 1 Outline. This work aims to provide a structured roadmap of the VLA paradigm for autonomous driving. We begin with **Preliminary Foundations** (Section 2), which formalize the general formulation of VLA models and detail their three core components: the multi-modal input modalities, the VLM backbone, and the action prediction head. It then traces the evolution from **VA Models** (Section 3), which directly map perception to control, towards **VLA Models** (Section 4), which incorporate language-grounded reasoning and interpretable decision-making. We further categorize VLA architectures into two major paradigms – **End-to-End VLA** (Section 4.1) and **Dual-System VLA** (Section 4.2) – that differ in their integration of vision, language, and action modules. Next, we review **Datasets & Benchmarks** (Section 5) that enable both open-loop and closed-loop evaluation of driving intelligence. Finally, we discuss **Challenges & Future Directions** (Section 6), highlighting interpretability, reasoning, and human-AI interaction as central themes driving the next generation of VLA-based autonomous driving research.

long-tailed scenarios [21–23]. Moreover, the sequential cascade is prone to cross-stage error propagation, where perception noise is amplified by downstream reasoning and control, compromising stability and safety.

To mitigate these issues, research has increasingly moved toward end-to-end autonomous driving, where **Vision-Action (VA)** models directly map raw sensory inputs to control commands or trajectory waypoints using imitation [24–26] and reinforcement learning [13, 27, 28]. Early systems such as ALVINN [29] and ChauffeurNet [30] demonstrated the viability of behavior cloning at scale. Subsequent advances introduced more expressive architectures: TransFuser [31] exploited transformer-based multimodal fusion, UniAD [32] unified perception and planning, VAD [33] leveraged vectorized scene representations, DriveTransformer [34] explored scalable transformer backbones, and DiffusionDrive [35] applied generative modeling to multi-modal trajectory prediction. Collectively, these VA models show that complex driving policies can be learned directly from data, laying the foundation for modern end-to-end AD systems.

Despite these successes, VA models exhibit fundamental limitations. They largely behave as "black boxes", offering limited interpretability in safety-critical settings [36–41]. Their generalization remains fragile under rare or long-tail scenarios that are underrepresented in training [36, 39, 42–45]. By directly mapping perception to low-level actions, they lack chain-of-thought (CoT) reasoning and contextual deliberation [36, 42, 46, 47], limiting their ability to resolve ambiguous or multi-stage interactions. Moreover, their focus on visual inputs

prevents them from incorporating high-level plans or human instructions in natural language, leaving a gap in human-vehicle interaction [21, 42, 48, 49].

The emergence of Large Language Models (LLMs) and Large Multimodal Models (LMMs) has catalyzed a new paradigm: **Vision-Language-Action (VLA)** models [50–52]. VLA models couple a Vision-Language Model (VLM) backbone with an action-prediction head, enabling direct mapping from multimodal inputs (vision + language) to executable driving actions. By jointly modeling perception, language understanding, and decision-making, VLA frameworks aspire to provide human-like reasoning, interpretability, and instruction-following [53–55]. Initial explorations such as DriveMLM [56] and GPT-Driver [57] introduced language modules into driving pipelines for high-level decision understanding, paving the way for more integrated designs. Later systems advanced toward closed-loop and reasoning-centric VLA models: LMDrive [49] achieved language-guided closed-loop driving, DriveLM [58] enabled structured reasoning via visual question answering, and DriveGPT4 [59] provided natural-language rationales for decisions. Recent works further investigate tightly coupled reasoning and control, including AutoVLA [53] with fast/slow thinking and GRPO-based optimization [60], and SimLingo [61], which explicitly studies language-action alignment.

End-to-End VLA models, however, must simultaneously *reason* and *act* in real time, creating challenges for latency and safety. This has led to *Dual-System* VLA designs, where high-level decision making is separated from low-level trajectory execution. DriveVLM [62] generates textual rationales or decisions with a VLM while relying on a classical planner for trajectories. VLP [21] tokenizes waypoints and value maps to produce planning-aware latent actions, and Diff-VLA [63] synthesizes language-guided trajectories refined by rule- or optimization-based controllers. InsightDrive [64] integrates causal language reasoning with MPC, assigning *why* to the VLM and *how* to the planner.

Together, these developments signal a paradigm shift from perception-driven pipelines toward systems that jointly reason, understand, and act. Given the rapid evolution of this field, there is a need to consolidate its conceptual foundations, clarify architectural trends, and provide a structured analysis of emerging directions.

Contributions. This work provides a comprehensive characterization of VLA models for autonomous driving. Specifically:

- We chart the evolution from precursor VA models (Section 3) to modern VLA frameworks (Section 4), providing historical context and clarifying the motivations behind this paradigm shift.
- We propose a taxonomy that categorizes VLA architectures into End-to-End (Section 4.1) and Dual-System (Section 4.2) designs, and compare their principles, advantages, and limitations.
- We present an organized synthesis of datasets and evaluation benchmarks relevant to VLA-based driving (Section 5), facilitating consistent and meaningful comparisons.
- We identify key challenges in real-world VLA deployment and outline future research directions (Section 6) to guide progress toward safer and more reliable autonomous systems.

Scope. This work differs from prior studies on VLA models [42, 50–52] and from concurrent surveys of planning-oriented end-to-end driving [65] through its domain-specific focus, historical framing, and architectural depth. [1]**Domain-specific focus.** Unlike broader analyses that span robotics or embodied AI [66, 67], our work *focuses exclusively on autonomous driving*, allowing a fine-grained analysis of driving-specific challenges, dataset characteristics, and safety requirements. [2]**Historical and conceptual continuity.** We adopt a "Past-Present-Future" narrative that traces the path from early VA models to modern VLA frameworks, emphasizing the motivations and technical lineage underlying the integration of language grounding into perception and control. [3]**Fine-grained architectural taxonomy.** Unlike prior high-level overviews [42, 68–70], we propose a hierarchical taxonomy that differentiates End-to-End and Dual-System VLA models and analyzes how they organize perception, reasoning, and control.

Through this combination of domain specificity, historical depth, and structured architectural analysis, we aim to provide a comprehensive and insightful reference for VLA research in autonomous driving.

The literature update covers works available through October 11, 2026. Recent preprints are discussed according to their reported architectures and evaluation settings; open-loop prediction, non-reactive simulation, and reactive closed-loop driving are distinguished when interpreting their evidence.



Figure 2 Summary of representative **VA** and **VLA** models from existing literature, spanning *End-to-End Models*, *World Models*, *Dual-Systems*, etc. For the complete list of related approaches and the discussions on their specifications, configurations, and technical details, kindly refer to Section 3 and Section 4, respectively.

Organization. The remainder of this paper is organized as follows. Section 2 introduces the preliminary foundations of VLA frameworks. Section 3 outlines the evolution of VA models. Section 4 presents our taxonomy and analysis of VLA architectures. Section 5 summarizes datasets and benchmarks. Section 6 discusses remaining challenges and future directions. Section 7 concludes this work.

2 Preliminary Foundations

Vision-Language-Action (VLA) frameworks [43, 53, 66, 71] leverage large Vision-Language Models (VLMs) [72–75] to interpret complex driving scenes and produce executable actions. A typical formulation can be expressed as:

$$\mathbf{a}_t = H(F(\mathbf{x}|\theta)) \ , \tag{1}$$

where $\mathbf{x}$ denotes multimodal inputs at timestamp t, $F(\cdot)$ is a VLM backbone parameterized by θ, and $H(\cdot)$ is an action-generation head. This section introduces these three components: the input modalities ($\mathbf{x}$), the VLM backbone (F), and the action prediction head (H).

2.1 Input Modalities

The input $\mathbf{x}$ aggregates heterogeneous signals that describe the external environment and the ego-vehicle state [76–78]. These inputs can be grouped into four categories: sensor observations, latent scene representations, language instructions, and proprioceptive states.

2.1.1 Sensor Inputs

Sensor inputs include raw or preprocessed data directly obtained from vehicle-mounted sensors [79–81].

- **Visual Images.** Surround-view RGB images that offer dense semantic information: $\mathbf{x}_{\text{img}} \in \mathbb{R}^{N_c \times H \times W \times 3}$, where N_c is the number of cameras (*e.g.*, 6 to 8), and H, W are the height and width of each image.
- **LiDAR Point Clouds.** A sparse or dense set of 3D points representing the environment geometry: $\mathbf{x}_{\text{lidar}} \in \mathbb{R}^{N_p \times D}, \quad D \geq 4$, where N_p is the number of points, and D includes dimensions such as x, y, z, velocity, and intensity.

2.1.2 Latent Representations

Multiple VLA systems operate on intermediate spatial representations that fuse multimodal sensor inputs.

- **Bird's-Eye View (BEV) Features.** Top-down view representation, often generated by fusing camera or LiDAR data [6, 82, 83]: $\mathbf{x}_{\text{bev}} \in \mathbb{R}^{C \times H_{\text{bev}} \times W_{\text{bev}}}$, where C is the number of feature channels, and $H_{\text{bev}}, W_{\text{bev}}$ are the spatial dimensions of the BEV grid.
- **Occupancy Grids.** 3D volumetric representation predicting occupancy and semantics for each spatial location [84–87]: $\mathbf{x}_{\text{occ}} \in \mathbb{R}^{C_{\text{occ}} \times X \times Y \times Z}$, where X, Y, Z are the spatial resolution of the 3D grid, and C_{occ} denotes the number of occupancy feature channels (*e.g.*, occupancy, flow and semantics).

2.1.3 Language Inputs

To enable VLA capabilities, the model also receives high-level textual instructions or task descriptions [59, 88, 89]. It is composed of a sequence of tokens representing the driving task or goal (*e.g.*, *"turn left at the next intersection"*): $\mathbf{x}_{\text{lang}} \in \mathbb{Z}^T$ or $\mathbf{x}_{\text{lang}} \in \mathbb{R}^{T \times D_{\text{emb}}}$, where T is the sequence length, and D_{emb} is the embedding dimension (if token embeddings are used).

2.1.4 Vehicle State Information

There is also proprioceptive information describing the current dynamic state of the ego-vehicle [76, 90]: $\mathbf{x}_{\text{state}} \in \mathbb{R}^{D_{\text{state}}}$, where D_{state} is the dimension of the state vector, including speed, acceleration, steering angle, yaw rate, turn indicator status, etc.

2.2 VLM Backbone (F)

The VLM backbone $F(\cdot)$ is the core reasoning engine of the system. It is typically a large vision language model [72, 74, 91–94]. Its primary role is to fuse the diverse input modalities into a single, powerful latent representation. It consists of a vision encoder (*e.g.*, a Vision Transformer, ViT) [95] to process visual inputs and an LLM decoder that conditions its generation on the visual features. A bridge network [74, 91] or unified multimodal token modelling mechanism [72, 92] is used to align the vision features with the language embeddings. VLM can directly generate the actions or provide the guidance for another action expert to develop more robust results.

2.2.1 VLM for Direct Action Generation (Single-System)

In this paradigm, the VLM directly emits actions through its language head [56–58, 96] or a small attached head [49, 61, 97]. This fully end-to-end design exploits the VLM's reasoning capabilities to map from visual/language inputs to executable controls.

2.2.2 VLM for Guidance Generation (Dual-System)

Alternatively, the VLM functions as a high-level reasoning module that produces intermediate guidance—textual rationales [62, 98, 99] or structured latent intents [21, 45], which a downstream planner converts into low-level actions [100]. This "slow thinking + fast execution" architecture improves interpretability and enables planners to enforce physical feasibility and safety constraints.

2.3 Action Prediction Head (H)

The head $H(\cdot)$ converts the VLM latent representation into action outputs. Consistent with the taxonomy used in existing literature, we categorize action heads into four types based on their output formulation and generation mechanism: Language Head (LH), Regression (REG), Trajectory Selection (SEL), and Trajectory Generation (GEN).

- **Language Head (LH).** This design directly utilizes the VLM's inherent text-generation capabilities to produce actions in the language space. The head is typically the language modeling head of the VLM, trained to output either free-form textual commands (*e.g.*, *"turn left"*) [59] or a sequence of discretized action tokens [53]. The model autoregressively predicts these tokens, which are subsequently parsed into executable signals. This approach is widely adopted in textual action generators like DriveMLM [56] and DriveGPT4 [59].
- **Regression (REG).** This formulation employs a decoder structure followed by a regressor (typically a Multi-Layer Perceptron) to directly predict continuous values. Unlike language heads, it avoids discretization by mapping the latent features aggregated via Transformers or GRUs to specific numerical outputs such as steering angles, throttle/brake values, or trajectory waypoints. Representative methods using this deterministic approach include LMDrive [49] and DriveGPT4-V2 [97].
- **Trajectory Selection (SEL).** Instead of directly regressing a single path, this head evaluates a set of candidate trajectories and selects the optimal one based on a learned cost function or scoring mechanism. The model typically generates or samples a diverse set of dynamically feasible trajectories and uses the latent representation to predict the cost or probability for each candidate. This approach, utilized by methods like WoTE [101] and SeerDrive [102], ensures that the final output adheres to kinematic constraints by selecting from pre-defined candidates.
- **Trajectory Generation (GEN).** This generative formulation synthesizes actions through probabilistic modeling, most notably using diffusion models or variational autoencoder [35, 63, 103]. Starting from noise, the head iteratively refines the trajectory sample conditioned on the VLM latent state and optionally language instructions. This allows the model to capture the multi-modality and uncertainty of future distributions. Prominent examples include ORION [54] and DiffVLA [63].

2.4 Action in Driving

In the context of autonomous driving, particularly for models like VLAs, the action space defines the set of possible outputs the model can generate to control the vehicle. The choice of action representation is a fundamental design decision that dictates how the model's reasoning is translated into physical motion. We outline three primary paradigms for action space representation below.

2.4.1 Discrete Trajectory Representations

This paradigm represents the vehicle's intended future path as a finite sequence of spatial waypoints [83]. Each waypoint is a spatial coordinate that the vehicle is expected to reach at a specific future time step. This representation allows for explicit geometric path planning and trajectory optimization. The action, $\mathbf{a}_t$, formulated at the current time t, is a set of Φ future waypoints:

$$\mathbf{a}_t = \{(x_i, y_i)\}_{i=1}^{\Phi}, \quad \text{where } (x_i, y_i) \in \mathbb{R}^2 \ . \tag{2}$$

Here, Φ is the prediction horizon (the total number of future steps), and each (x_i, y_i) is a coordinate in a 2D Cartesian plane representing the target position at step i.

2.4.2 Continuous Trajectory Representations

Instead of discrete points, this approach parameterizes the vehicle's motion as a continuous function over a future time horizon [104]. The trajectory is typically defined by functions that govern the vehicle's longitudinal and lateral motion, such as speed and turning radius. The action, a_t, is defined by these continuous functions

over a time interval $[0, T]$:

$$\mathbf{a}_t = (v(t), \kappa(t)), \quad \text{for } t \in [0, T] \ . \tag{3}$$

In this formulation, $v(t)$ represents the vehicle's speed profile, and $\kappa(t)$ represents its curvature profile over the future time horizon T. This inherently captures the continuous nature of vehicle dynamics.

2.4.3 Direct Control Representations

This paradigm involves the direct output of low-level vehicle control commands that are immediately sent to the vehicle's actuators [97]. These outputs typically consist of continuous signals for steering, acceleration, and braking control. The values are often normalized and constrained to lie within the vehicle's physical operational ranges. The action vector, a_t, represents control signals for a specific time step t:

$$\mathbf{a}_t = (\delta_t, \tau_t, \beta_t) \ , \tag{4}$$

where δ_t is the steering angle, τ_t is the throttle input, and β_t is the brake input at time step t. Each component is bound by the vehicle's hardware limits, *e.g.*, $\delta_t \in [\delta_{\min}, \delta_{\max}]$.

2.4.4 Language Representations

This paradigm leverages the natural language capabilities of VLMs to express driving actions through textual descriptions [105]. The action is represented as a sequence of discrete tokens from a predefined vocabulary:

$$\mathbf{a}_t = \{w_1, w_2, \ldots, w_T\}, \quad \text{where } w_i \in \mathcal{V} \ . \tag{5}$$

Here, $\mathcal{V}$ represents the model's vocabulary, T is the sequence length, and each token w_i corresponds to an element in the vocabulary. The language-based action can range from high-level commands (*e.g.*, *"turn left at the intersection"*) to specific numerical trajectory representations encoded as text tokens.

3 Vision-Action Models

Vision-Action (VA) models represent one of the earliest and most influential lines of research in autonomous driving. Their core idea is to directly map sensory observations – typically camera inputs – to driving actions, thereby avoiding explicit modular decomposition into perception, prediction, and planning. Enabled by deep neural networks, VA models have been explored through **two major training paradigms**: [1]*imitation learning*, which distills policies from expert demonstrations, and [2]*reinforcement learning*, which optimizes behavior through trial-and-error interaction. More recently, *world models* have expanded this paradigm by enabling agents to simulate scene dynamics and reason about action consequences, improving robustness and scalability. Table 1 provides an overview of representative efforts.

From an architectural perspective, VA methodologies for autonomous driving can be broadly grouped into:

- **End-to-End Models**, which directly predict control commands or planned trajectories from sensory inputs.
- **World Models**, which explicitly model action-conditioned future dynamics to support policy learning and decision-making.

3.1 End-to-End Models for Autonomous Driving

End-to-end (E2E) models learn a single neural network that maps raw or intermediate sensor observations to actions or planned trajectories [2, 36, 101, 106]. Unlike modular stacks, which isolate perception, prediction, and planning, E2E approaches implicitly couple these tasks within a unified representation [107–109]. Depending on whether perception supervision is employed, existing methods fall into two main categories: *action-only models* and *perception-action models*, as illustrated in Figure 3.

3.1.1 Action-Only Model

Action-only models adopt a streamlined one-stage formulation: sensory inputs are fed directly into a network that outputs low-level actions. These methods primarily differ in whether policies are learned from **demonstrations** or through **exploration**.

Imitation Learning (IL), especially behavior cloning [174], learns a policy by matching expert actions, as visualized in the IL branch of Figure 3. Early works [24, 25, 29, 106] demonstrated that actions can be predicted directly from monocular or multi-view inputs, and subsequent designs refined backbone architectures [31, 80, 175].

NEAT [123] highlights behaviorally relevant image regions via intermediate attention maps, while TCP [126] fuses a trajectory branch and control branch for complementary supervision. To better leverage scene geometry, BEV-Planner [90] predicts trajectories from BEV features enriched with ego states. Urban-Driver [127] moves beyond open-loop evaluation by training policies in a differentiable, data-driven simulator.

IL-based methods are simple, efficient, and require no reward engineering; however, they remain sensitive to distribution shift [110, 176, 177] and causal confusion [178–180], which can impair reliability in long-tailed or rare-event scenarios.

Reinforcement Learning (RL) optimizes actions through interaction, offering greater flexibility than imitation-based approaches [181, 182]. Several works address the sample inefficiency of RL by combining it with supervised pretraining: Latent-DRL [122] and Gri [128] pre-train visual encoders using semantic segmentation, while LSD [183] initializes policies via IL before performing RL fine-tuning. Privileged-information distillation has also proven effective: LBC [79], WoR [125], and Roach [124] use simulator-only states to guide sensor-based agents.

Combined with the world model,

Table 1 Summary of **Vision-Action** models in autonomous driving. ViRA uses frozen foundation-model teachers only during training; its encoder and action head follow the evaluated VA planner.
- **Inputs**: Camera, LiDAR, and Ego-Status.
- **Action Types**: RL: Policy w/ Reinforcement Learning, REG: Decoder + MLP, SEL: Traj. Selection w/ Cost, and GEN: Traj. Generation w/ Generative Model.
- **Outputs**: Ctrl.: Control Signal, Traj.: Numerical Trajectory.
- **Datasets:** CA CARLA [110], NC NoCrash [111], PG ProcGen [112], LY Lyft [113], NS nuScenes [76], B2 Bench2Drive [114], NV NAVSIM [115], OO OpenOcc [116], OV OpenDV [117], NP nuPlan [118], O3 Occ3D [119], CO Cam4DOcc [120], and PR Private Data.

#	Model	Venue	Input	Dataset	Vision	Action	Output
• Sec. 3.1.1 Action-only Models							
1	LBC [79]	CoRL'20		CA NC	ResNet [121]	RL	Ctrl.+Traj.
2	Latent-DRL [122]	CVPR'20		CA	ResNet [121]	RL	Ctrl.
3	NEAT [123]	ICCV'21		CA	ResNet [121]	REG	Traj.
4	Roach [124]	ICCV'21		CA NC	ResNet [121]	RL	Ctrl.
5	WoR [125]	ICCV'21		CA NC PG	ResNet [121]	REG	Ctrl.
6	TCP [126]	NeurIPS'22		CA	ResNet [121]	REG	Ctrl.+Traj.
7	Urban-Driver [127]	CoRL'22		LY	ResNet [121]	REG	Traj.
8	LAV [80]	CVPR'22		CA	ResNet [121]	REG	Ctrl.+Traj.
9	TransFuser [31]	TPAMI'23		CA	ResNet [121]	REG	Traj.
10	GRI [128]	Robotics'23		CA	EfficientNet [129]	RL	Ctrl.
11	BEVPlanner [90]	CVPR'24		NS	ResNet [121]	REG	Traj.
12	Raw2Drive [130]	NeurIPS'25		CA B2	ResNet [121]	RL	Ctrl.
13	RAD [28]	NeurIPS'25		PR	ResNet [121]	RL	Traj.
14	TrajDiff [131]	arXiv'25		NV	ResNet [121]	GEN	Traj.
• Sec. 3.1.2 Perception-Action Models							
15	ST-P3 [109]	ECCV'22		NS CA	EfficientNet [129]	SEL	Traj.
16	UniAD [32]	CVPR'23		NS	ResNet [121]	REG	Traj.
17	VAD [33]	ICCV'23		NS	ResNet [121]	REG	Traj.
18	OccNet [116]	ICCV'23		NS OO	ResNet [121]	SEL	Traj.
19	GenAD [103]	ECCV'24		NS	ResNet [121]	GEN	Traj.
20	PARA-Drive [132]	CVPR'24		NS	ResNet [121]	REG	Traj.
21	Hydra-MDP [133]	CVPRW'24		NV	ResNet [121]	SEL	Traj.
22	SparseAD [134]	arXiv'24		NS	ResNet [121]	REG	Traj.
23	GaussianAD [135]	arXiv'24		NS	ResNet [121]	REG	Traj.
24	DiFSD [136]	arXiv'24		NS	ResNet [121]	GEN	Traj.
25	DriveTransformer [34]	ICLR'25		NS B2	ResNet [121]	REG	Traj.
26	SparseDrive [107]	ICRA'25		NS	ResNet [121]	REG	Traj.
27	DiffusionDrive [35]	CVPR'25		NS NV	ResNet [121]	GEN	Traj.
28	GoalFlow [137]	CVPR'25		NV	VoVNet [138]	GEN	Traj.
29	GuideFlow [139]	arXiv'25		NS NV B2	ResNet [121]	GEN	Traj.
30	ETA [140]	arXiv'25		B2	CLIP-ViT [95]	REG	Traj.
31	Geo [141]	arXiv'25		NS	ResNet [121]	REG	Traj.
32	DiffusionDriveV2 [142]	arXiv'25		NV	ResNet [121]	GEN	Traj.
33	NaviHydra [143]	arXiv'25		NV	ResNet [121]	SEL	Traj.
34	Mimir [144]	arXiv'25		NV	ResNet [121]	GEN	Traj.
35	ViRA [145]	arXiv'26		NV (v2)	ResNet-34	REG/GEN/SEL	Traj.
• Sec. 3.2.1 Image-Based World Models							
36	DriveDreamer [146]	ECCV'24		NS	SD [147]	REG	Traj.
37	GenAD [117]	CVPR'24		OV	SDXL [148]	REG	Traj.
38	Drive-WM [149]	CVPR'24		NS	ConvNeXt [150]	SEL	Traj.
39	DrivingWorld [151]	arXiv'24		NP	VQ-VAE [152]	REG	Traj.
40	Imagine-2-Drive [153]	IROS'25		CA	SVD [154]	SEL	Traj.
41	DrivingGPT [155]	ICCV'25		NP NV	VQ-VAE [152]	REG	Traj.
42	Epona [156]	ICCV'25		NS NV NP	DC-AE [157]	REG	Traj.
43	VaViM [158]	arXiv'25		OV NS NP	LLaMAGen [159]	GEN	Traj.
• Sec. 3.2.2 Occupancy-Based World Models							
44	OccWorld [160]	ECCV'24		NS O3	ResNet [121]	REG	Traj.
45	NeMo [161]	ECCV'24		NS	ResNet [121]	REG	Traj.
46	OccVAR [162]	-		NS O3	ResNet [121]	REG	Traj.
47	RenderWorld [163]	arXiv'24		NS O3	Swin-T [164]	REG	Traj.
48	DFIT-OccWorld [165]	arXiv'24		NS O3	ResNet [121]	REG	Traj.
49	Drive-OccWorld [166]	AAAI'25		NS CO	ResNet [121]	REG	Traj.
50	T^3Former [167]	arXiv'25		NS O3	ResNet [121]	REG	Traj.
51	AD-R1 [168]	arXiv'25		NS NV	-	RL	Traj.
• Sec. 3.2.3 Latent-Based World Models							
52	Covariate-Shift [26]	arXiv'24		CA	DINOv2 [169]	REG	Traj.
53	World4Drive [170]	ICCV'25		NS NV	ResNet [121]	REG	Traj.
54	WoTE [101]	ICCV'25		NV B2	ResNet [121]	SEL	Traj.
55	LAW [171]	ICLR'25		NS NV CA	Swin-T [164]	REG	Traj.
56	SSR [172]	ICLR'25		NS CA	ResNet [121]	REG	Traj.
57	Echo-Planning [173]	arXiv'25		NS	ResNet [121]	REG	Traj.
58	SeerDrive [102]	NeurIPS'25		NS NV	VoVNet [138]	SEL	Traj.

Think2Drive [184] trains the agent
with the Model-Based RL (MBRL) method, paired with a compact latent world model learning the transitions of the environment. Raw2Drive [130] is a dual-stream MBRL approach, where the raw sensor world model is aligned with the privileged world model for camera-based action prediction. In contrast to studies in non-photorealistic CARLA [110], recent efforts have shifted toward photorealistic world modeling. RAD [28] establishes a 3DGS-based [185] closed-loop RL training paradigm regulated by IL in a realistic 3DGS environment. The key challenges in RL-based models include sample inefficiency [13], reward function design [186], and sim-to-real transfer [2].

Temporal Memory and Rollout-Aware Post-Training. Two limitations of the imitation recipe drew targeted attention this window. The first is that historical context is treated as uniformly useful: Velocity-Aware Selective Memory [187] argues that not all history helps and conditions retention on ego velocity for long-horizon planning, while MomADv2 [188] filters historical planning queries so that memory which has become invalid — typically after the driving command changes — cannot mislead the current decision. The second is that end-to-end policies remain predominantly trained on fixed expert data with open-loop imitation even when evaluated closed-loop; RoG-DAgger [189] addresses this with rollout-guided post-training, revisiting dataset aggregation for modern driving stacks. Curriculum construction is being reframed in the same spirit: CL4AD [190] treats scenario selection in batched simulators as unsupervised environment design rather than a fixed schedule, shaping curricula from success rates.

3.1.2 Perception-Action Model

Perception-action models follow a two-stage paradigm in which perception tasks (*e.g.*, mapping, tracking) supervise and constrain trajectory prediction. These methods generally adopt either dense BEV-based representations or sparse query-based representations, as shown in Figure 3.

Dense BEV-Based Models construct unified top-down features from multi-view cameras. ST-P3 [109] jointly learns spatial-temporal features for perception and planning; UniAD [32] integrates sequential task dependencies to support goal-directed planning. VAD [33] employs a vectorized scene representation to improve both planning safety and efficiency. OccNet [116] incorporates occupancy embeddings to capture 3D scene geometry. Para-Drive [132] proposes a fully parallel E2E architecture for real-time deployment. Addressing partial observability, BeyondSight [191] maintains object permanence for actors that become fully occluded, and a complementary VLM-based approach [192] explicitly identifies which occluded agents are planning-critical rather than treating all occlusions with uniform conservatism. Plug-and-Play Traffic Element Awareness [193] similarly augments E2E planners with dedicated traffic-light and road-sign perception, a role historically underweighted relative to dynamic agents.

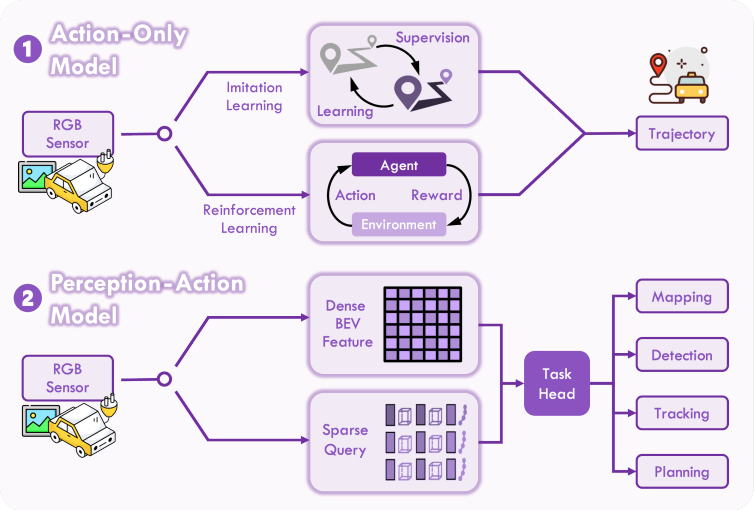


Figure 3 The categorization of **End-to-End VA models** based on model structures and outputs, including *Action-Only Models* (Sec. 3.1.1), and *Perception-Action Models* (Sec. 3.1.2).

Generative and sampling-based approaches have recently emerged [194]: GenAD [103] frames planning as sampling from learned distributions; DiffusionDrive [35] introduces a truncated diffusion policy guided by multi-modal anchors; GuideFlow [139] incorporates explicit physical constraints into the generation process. These generative planners are not immune to attack: Off the Rails [195] shows that the scoring head shared by diffusion- and retrieval-based generative E2E planners can be hijacked with safety-critical consequences, exposing a new attack surface specific to this design family. While BEV representations naturally align with 2D trajectory planning, they require substantial computation due to their dense spatial structure.

Sparse Query-Based Models avoid explicit BEV grids by using latent queries to aggregate image features. SparseAD [134] and SparseDrive [107] represent the entire scene using sparse perception queries and a parallel planner, achieving strong efficiency-accuracy trade-offs. DiFSD [136] introduces an ego-centric sparse formulation and models uncertainty through trajectory denoising. DriveTransformer [34] incorporates task parallelism, sparse attention, and streaming updates for improved stability. GaussianAD [135] adopts 3D semantic Gaussians for fine-grained yet compact scene representation.

Sparse query methods significantly reduce inference latency, but the absence of a dense future-world representation can restrict long-horizon reasoning and planning safety.

ViRA [145] examines a complementary training interface: frozen visual foundation models supervise spatial and scene-semantic representation alignment across several VA planners, while the deployed planner remains unchanged. Its findings connect the choice of teacher representation to auxiliary perception supervision; they do not establish equivalent gains for VLA models.

3.2 World Models for Autonomous Driving

World models aim to predict how driving scenes evolve under different ego actions [196–198]. By jointly modeling scene dynamics and ego motion, they provide a powerful mechanism for learning safe, long-horizon driving policies [199]; recent surveys further consolidate this rapidly growing literature into unified taxonomies and evaluation frameworks [200, 201], while recent analyses interrogate the paradigm's causal assumptions, asking specifically how a driving world model should support counterfactual prediction [202]. Their applications span immersive simulation [203–208], end-to-end planning [44, 146, 149, 151], and feature learning for downstream tasks [209–215]. Here, we focus on world models designed for trajectory planning and categorize them by prediction modality and representation granularity into three groups: **image-based**, **occupancy-based**, and **latent-based** models (Figure 4).

3.2.1 Image-Based World Model

Image-based world models generate future frames conditioned on ego actions, enabling agents to "dream" scene evolution and evaluate the consequences of different trajectories. These methods leverage modern generative models to synthesize realistic, temporally coherent videos [216] and are typically classified into diffusion-based and autoregressive architectures.

Diffusion-Based World Models use latent video diffusion [147, 217] to produce multi-step rollouts. For front-view forecasting, GenAD [117] and Vista [44] incorporate temporal reasoning modules to handle complex motion patterns. Imagine-2-Drive [153] integrates diffusion generation into a reinforcement-learning framework, training a policy actor inside the world model. To support multi-view predictions, DriveDreamer [146] employs a two-stage pipeline for video synthesis and policy learning. Drive-WM [149] factors views within a spatiotemporal model and generates multiple plausible futures, selecting trajectories using image-based rewards. Pushing toward scalable and controllable multi-camera generation, X-World [218] learns ego-centric multi-camera world models for end-to-end driving, while the Xiaomi Auto World Model [219] jointly couples reconstruction and generation within a single driving world model. Targeting closed-loop deployment, OmniDreams [220] provides a real-time generative world model for closed-loop autonomous-vehicle simulation. Orbis 2 [221] organizes rollout generation hierarchically for long-horizon stability, and M^{4}World [222] extends multi-view generation with interactive object-level control. On the sampling side, a training-free norm-injection technique [223] asks whether explicit energy guidance is even necessary for controllable rollouts, and GeoRoute [224] targets the geometry-drift and temporal-ghosting artifacts that plague long-horizon future-frame prediction more broadly, across both driving and traffic-surveillance settings. Beyond forward simulation, driving world models are increasingly used as data engines: Cam2Sim [225] reconstructs closed-loop simulation scenarios directly from camera footage, RealWeather [226] translates driving footage across weather conditions while preserving scene fidelity, and SUV [227] reframes future scene understanding itself as a video-generation task for downstream E2E planning.

Autoregressive (AR) Models tokenize images using VQ-VAE [152] and model scene evolution via next-token prediction [228–230]. DrivingWorld [151] builds a GPT-style architecture for high-fidelity long-horizon video generation. DrivingGPT [155] interleaves image and action tokens, unifying simulation and trajectory planning.

Epona [156] combines AR modeling with diffusion to produce high-resolution, long-duration rollouts, and EponaV2 [231] extends this design with comprehensive future reasoning to better serve prediction and planning.

Image-based world models provide photorealistic simulations crucial for training and evaluation. However, their reliance on 2D appearance limits explicit 3D reasoning, which can hinder safety-critical long-horizon planning.

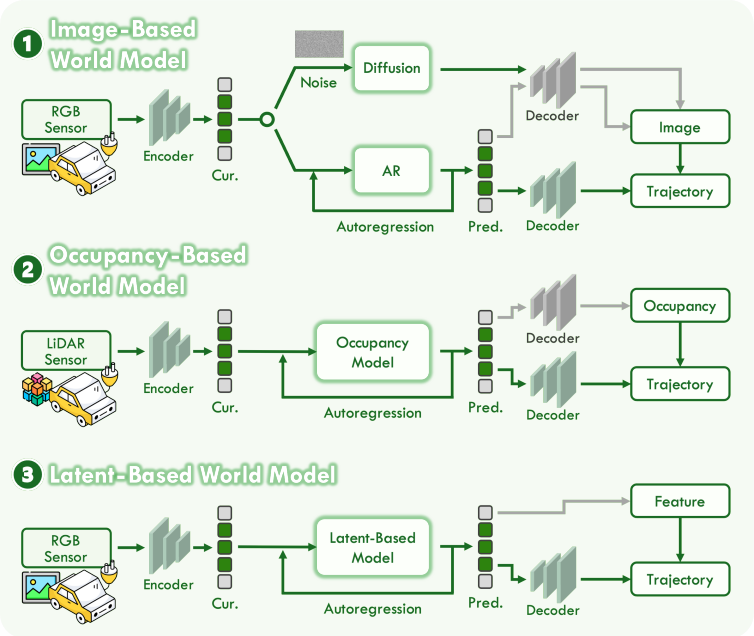


Figure 4 The categorization of **World Models** based on prediction modalities, including *Image-Based Models* (Sec. 3.2.1), *Occupancy-Based Models* (Sec. 3.2.2), and *Latent-Based Models* (Sec. 3.2.3).

3.2.2 Occupancy-Based World Models

Occupancy-based world models represent the driving scene as spatiotemporal occupancy grids and predict their evolution under different actions [232]. Instead of synthesizing raw pixels, these models focus on the geometry and semantics of free space, obstacles, and agents [208, 233–235]. As shown in the middle of Figure 4, AR prediction is commonly used in occupancy world models.

OccWorld [160] first introduces occupancy forecasting for planning, using a scene tokenizer to discretize 3D occupancy before applying a GPT-style transformer to synthesize future scenes and ego trajectories. RenderWorld [163] produces 3D occupancy through a self-supervised Gaussian module, while OccVAR [162] performs coarse-to-fine 4D occupancy forecasting. T^{3}Former [167] encodes occupancy using compact triplanes and predicts future triplane updates from multi-scale history.

An alternative line employs single-stage feedforward prediction. Drive-OccWorld [166] uses predicted future BEV features for action-conditioned 3D forecasting. DFIT-OccWorld [165] introduces a decoupled dynamic flow strategy to support efficient non-autoregressive prediction. NeMo [161] improves vision-based occupancy forecasting by combining it with self-supervised image reconstruction signals. GEM [236] casts occupancy forecasting as Gaussian evolution and jointly optimizes 4D occupancy prediction with motion planning. OWMDrive [237] instead foregrounds causality, using a 4D occupancy world model to support causality-aware E2E planning, and GaussianDWM++ [238] grounds the Gaussian occupancy representation in language, coupling unified scene understanding with controllable 4D editing.

Occupancy-based models offer strong geometric fidelity and explicit free-space reasoning but rely on costly 3D annotations, which can limit scalability across diverse environments.

3.2.3 Latent-Based World Models

Latent-based world models bypass explicit image or occupancy forecasting and instead predict future dynamics directly in a compressed latent space [239–241]. By operating on high-level features, these models capture behavioral patterns and long-horizon dependencies while avoiding the computational overhead of pixel-level generation. Regarding the forecasting strategy, the latent world model utilizes single-frame or AR prediction presented at the bottom of Figure 4.

Early latent models [242, 243] learn feature-level dynamics for planning without generating visual frames. LAW [171] leverages self-supervised learning to predict future scene features from current features and planned ego trajectories, enabling end-to-end driving without perception labels. World4Drive [170] employs vision foundation models to create latent representations from which diverse planning trajectories can be generated

and evaluated. Echo-Planning [173] introduces a bidirectional Current→Future→Current (CFC) cycle to enforce temporal consistency in latent BEV features. For robustness in imitation learning, Covariate-Shift [26] addresses distribution mismatch using latent rollouts. Two recent JEPA-style designs push latent pretraining further: CRISP [244] pretrains a spatiotemporal camera-radar backbone via forecasting-based objectives, while a geographic-diversity study [245] shows that broadening the geographic spread of a zero-label JEPA driving corpus improves cross-domain generalization more than simply adding volume; Auto-JEPA [246] builds directly on this line, learning a latent world model of continuous driving intent for end-to-end planning. By injecting predicted BEV features, SeerDrive [102] refines both latent prediction and trajectory generation in a closed-loop manner. Several recent designs make the latent representation more efficient or controllable: SparseWorld [247] adopts a sparse scene representation to enhance end-to-end planning, Unified Driving Tokens [248] learns a representation- and geometry-guided discrete tokenizer that couples world modeling with planning, and Kinematics-Aware latent world models [249] inject vehicle kinematics for data-efficient policy learning. Other works re-frame the forecasting objective itself: IDOL [250] guides end-to-end planning through inverse-dynamics-based future prediction, while DynVLA [251] learns world dynamics to support action reasoning.

A growing line of work unifies generative world modeling with planning under a shared latent space. World-action formulations such as Metis [252] couple video-world generation with action denoising, and several methods bridge scene generation and decision-making by unifying vision and motion representations [253, 254]. HERMES++ [255] targets a unified driving world model for joint 3D scene understanding and generation, GraphWorld [256] introduces a long-horizon graph-structured world model for end-to-end planning, and DriveVA [257] shows that video-action models can act as zero-shot drivers. DVGT-2 [258] provides a vision-geometry-action alternative that augments visual features with explicit geometry without a language model. Building directly on the VLA paradigm, DriveWorld-VLA [259] performs unified latent-space world modeling jointly with vision-language-action reasoning, while ExploreVLA [260] combines dense world modeling with exploration for end-to-end driving. Extending this convergence, BrainWAM [261] explicitly coordinates semantic priors with predictive dynamics in a shared action space, X-Mind [262] distills a predictive world model into an efficient visual chain-of-thought for VLA reasoning, WCog-VLA [263] couples world-cognitive dual-level representations with end-to-end VLA planning, HyWorldVLA [264] hybridizes world modeling within the VLA reasoning loop, and LWDrive [265] guides VLM-based planning layer by layer with world-model feedback to sharpen coarse language-derived intentions into geometrically accurate trajectories. As these systems grow more capable, their inference cost also grows; DriveCache [266] addresses this directly with action-aware caching that accelerates driving world-model inference without retraining.

Latent world models offer efficient and semantically informed forecasting. However, achieving high-quality planning still requires auxiliary supervision from 2D/3D annotations, such as bounding boxes or HD maps.

Making Latent Prediction Decision-Relevant. A recurring criticism of latent driving world models is that their futures are predictive but not *decision-informative*. DA-WAM [267] makes this precise: existing approaches either decouple future-representation learning from planning optimization or share one predicted state across all candidate trajectories, which dilutes exactly the action-specific consequences that should discriminate between them; it therefore aligns the learned latents with trajectory selection. WA-JEPA [268] attacks the same gap at the pretraining objective, arguing that V-JEPA's random-mask completion and deterministic regression are ill-suited to planning, and substituting future-masked, action-coupled prediction. GeoWAM [269] moves the prediction target out of pixel space into visual geometry, reporting stronger open- and closed-loop policies than image-based world modeling, and RISE [270] observes that a fixed imagination budget is wasteful and allocates rollout effort per scene. Together these mark a shift from asking whether a world model predicts well to asking whether its predictions change the chosen action.

Recent world-action models further tighten the connection between imagined futures and executable plans. PhysWAM [271] jointly generates multi-view video, metric depth, and ego motion, using geometric supervision during training and a parameter-free consensus mechanism for trajectory selection at inference. ReWAM [272] instead represents ego actions and surrounding-agent responses through level-k reasoning over shared future latents. World4Scorer [273] learns candidate-conditioned future representations and outcome-based scoring, anchoring the executed candidate to observed futures while supervising alternative candidates with simulated outcomes. These designs emphasize geometry, interaction, and decision relevance, respectively; their predictive structure should be assessed separately from the reactivity of the evaluation environment.

Table 2 Categories of **natural language prompts** for Vision-Language-Action (VLA) models in autonomous driving.

Prompt Type	Explanations
System Prompt	Text templates or query formulations designed to interact with LLMs, guiding them to perform specific driving-related reasoning or trajectory prediction tasks. System prompts often define the task structure, provide role definitions, and shape the model's reasoning behavior.
Instructions	Commands or instructions provided by humans or systems, typically describing the driving goal or required maneuver (*e.g.*, "turn left at the next intersection").
Scene Description	Textual descriptions of surroundings, including perceived objects, road layout, and contextual factors (*e.g.*, *"a pedestrian is crossing on the right"*, or *"a vehicle is 5 meters ahead on the left"*).
Traffic Rules	Prompts encoding regulatory constraints or domain knowledge, such as traffic laws, traffic light status, right-of-way rules, or safety guidelines.
Ego Status	Information about the ego vehicle's internal state, including speed, position, heading, or navigation intent.
Context Information	Demonstrations presented as paired examples of driving scenarios and corresponding actions, used to guide the model via in-context learning.

3.3 Limitations of VA Compared to VLA

While VA models remain widely deployed, they face structural limitations that hinder performance in complex, ambiguous, or long-tailed scenarios: areas where VLA models excel. Some recent VA systems attempt to close this gap without adopting a full VLA stack, for instance by injecting explicit traffic-rule constraints as neuro-symbolic safety guards over an otherwise black-box E2E planner [274]; nonetheless, the limitations below remain characteristic of the broader VA paradigm.

- **Limited Interpretability.** VA models provide little insight into their decision-making process. In contrast, VLA models can articulate reasoning steps or explanations through language.
- **Weak Generalization.** VA policies lack broad world knowledge and often require environment-specific retraining. VLA models leverage large-scale pretraining to generalize better under distribution shifts and long-tailed events.
- **No Chain-of-Thought Reasoning.** VA models directly map pixels to actions, making it difficult to perform explicit reasoning or contextual analysis. VLAs natively support step-wise reasoning.
- **No Language Understanding.** VA systems cannot incorporate human instructions or high-level goals expressed in texts. VLA models naturally integrate such inputs to guide planning and decision-making.

4 Vision-Language-Action Models

VLA models extend the Vision-Action paradigm by coupling visual perception with the multimodal reasoning capabilities of large vision-language models. Equipped with chain-of-thought style inference and broad world knowledge, these models are particularly promising for rare, ambiguous, and long-tailed driving scenarios. Table 2 summarizes typical prompting strategies, and Table 3 overviews representative VLA-based approaches.

Table 3 Summary of **Vision-Language-Action** models in autonomous driving.
• **Input**: 📷:Camera, 💬: Sys. Prompt, 🔶: Instruct., 🗺️: Scene Descrip., 📈: Status, 🚦: Traffic Rule, 📋: Context Info.
• **Action**: LH: Lang. Head, REG: Decoder+MLP, GEN: Traj. GEN with Generative Model.
• **Output**: Desc.: Linguistic Descriptions, Traj.: Numerical Trajectory, Ctrl.: Control Signal. Meta.: Meta Action.
• **Datasets:** NS: nuScenes [76], BX: BDD-X [275], DL: DriveLM [58], SD: SDN [276], VA: VLAAD [277], B2: Bench2Drive [114], WA: Waymo [278], MA: MetaAD [279], CA: Carla [110], IM: ImpromptuVLA [280], NV: NAVSIM [115], OV: OpenDV [117], NP: nuPlan [118], TC: Talk2Car [281], CV: CoVLA [43], PA: PhysicalAI-AV [282] and PR: Private Data.

#	Model	Venue	Input Modality	Dataset	Vision	Language	Action	Output
• Sec. 4.1.1: Textural Action Generator								
1	DriveMLM [56]	arXiv'23	📷💬🔶🚦	CA	EVA-CLIP [283]	LLaMA [284]	LH	Desc.+Meta.
2	RAG-Driver [39]	RSS'24	📷💬🔶📋📈	BX	CLIP [95]	Vicuna-1.5 [285]	LH	Desc.+Ctrl.
3	RDA-Driver [286]	ECCV'24	📷💬📈	DL NS	BEVFusion [82]	LLaMA [284]	LH	Desc.+Traj.
4	DriveLM [58]	ECCV'24	📷💬	DL	BLIP-2 [287]	LLaMA [284]	LH	Meta.+Traj.
5	DriveGPT4 [59]	RA-L'24	📷💬	BX	CLIP [95]	LLaMA-2 [284]	LH	Desc.+Ctrl.
6	DriVLMe [288]	IROS'24	📷💬🔶	BX SD CA	CLIP [95]	Vicuna [285]	LH	Desc.+Ctrl.
7	LLaDA [289]	CVPR'24	📷💬🔶🗺️🚦	NS NP	-	GPT-4 [290]	LH	Ctrl.+Traj.
8	VLAAD [277]	WACVW'24	📷💬🔶	VA	BLIP-2 [287]	LLaMA-2 [284]	LH	Ctrl.
9	OccLLaMA [85]	arXiv'24	📷💬	NS	Swin-T [164], PointPillar [291]	LLaMA-3 [292]	LH	Ctrl.+Traj.
10	Doe-1 [293]	arXiv'24	📷💬📈🗺️	NS	Lumina-mGPT [294]	BPE tokenizer [295]	LH	Ctrl.+Traj.
11	LINGO-2 [296]	arXiv'24	📷💬🔶	PR	Wayve Vision Model [297]	Wayve VLA Model [297]	LH	Desc.+Traj.
12	SafeAuto [48]	ICML'25	📷💬📈🚦📋	BX DL	LanguageBind [298]	Video-LLaVA [299]	LH	Desc.+Ctrl.
13	OpenEMMA [300]	WACV'25	📷💬📈	NS	Qwen2-VL [72]	Qwen2-VL [72]	LH	Desc.+Traj.
14	ReasonPlan [301]	CoRL'25	📷💬🔶📈	B2	SigLIP [302]	Qwen [303]	LH	Desc.+Traj.+Meta.
15	FutureSightDrive [304]	NeurIPS'25	📷💬🔶📈	NS DL	ViT [305]	Qwen2-VL [72]	LH	Desc.+Traj.
16	ImpromptuVLA [280]	NeurIPS'25	📷💬🗺️📈🚦	IM	Qwen2.5-VL [92]	Qwen2.5-VL [92]	LH	Traj.
17	WKER [306]	AAAI'25	📷💬🔶📈	NS	EVA-02 [307]	LLaMA3 [292]	LH	Desc.+Traj.
18	OmniDrive [308]	CVPR'25	📷💬🔶🗺️📈🚦	NS DL	EVA-02 [307]	LLaMA2 [284]	LH	Desc.+Traj
19	S4-Driver [309]	CVPR'25	📷💬🔶📈	WA NS	PaLI3 [310]	PaLI3 [310]	LH	Meta.+Traj.
20	EMMA [96]	TMLR'25	📷💬🔶📈	WA NS	Gemini-VLM [311]	Gemini [312]	LH	Desc.+Traj.
21	Occ-LLM [86]	ICRA'25	📷💬📈	NS	-	LLaMA2 [284]	LH	Traj.
22	Sce2DriveX [313]	RA-L'25	📷💬🔶📈	NS B2	OpenCLIP [314]	Vicuna-1.5 [285]	LH	Ctrl.+Traj.
23	DriveAgent-R1 [315]	arXiv'25	📷💬🔶📈	NS PR	Qwen2.5-VL [92]	Qwen2.5-VL [92]	LH	Desc.+Meta.
24	Drive-R1 [316]	arXiv'25	📷💬🔶📈	NS DL	InternVL2 [317]	InternVL2 [317]	LH	Desc.+Traj.+Meta.
25	FastDriveVLA [105]	arXiv'25	📷💬	NS	Qwen2.5-VL [92]	Qwen2.5-VL [92]	LH	Desc.+Traj.
26	WiseAD [318]	arXiv'25	📷💬	BX CA DL	CLIP [95]	MobileVLM [75]	LH	Traj.+Ctrl.
27	AutoDrive-R² [319]	arXiv'25	📷💬📈	NS WA	Qwen2.5-VL [92]	Qwen2.5-VL [92]	LH	Traj.
28	OmniReason [320]	arXiv'25	📷💬🔶🗺️📈🚦📋	NS B2	EVA-02 [307]	LLaVA-1.5 [321]	LH	Meta.+Traj.
29	OpenREAD [322]	arXiv'25	📷💬🔶🗺️📈	NS	Qwen3 [323]	Qwen3 [323]	LH	Desc.+Meta.+Traj.
30	dVLM-AD [324]	arxiv'25	📷💬📈	WA NS	SigLIP2 [325]	LLaDA-V [326]	LH	Desc.+Traj.
31	PLA [327]	arXiv'25	📷💬🔶🗺️📈📋	NS	Sensor Encoder	GPT-4.1 [290]	LH	Desc.+Traj.
32	AlphaDrive [279]	arXiv'25	📷💬🔶📈	MA	Qwen2-VL [72]	Qwen2-VL [72]	LH	Desc.+Meta.
33	CoReVLA [328]	arXiv'25	📷💬🔶📋	B2	Qwen2.5-VL [92]	Qwen2.5-VL [92]	LH	Ctrl.+Traj.
34	FocusDrive [329]	arXiv'26	📷💬🔶📈	NV	Qwen3-VL ViT	Qwen3-VL-2B	LH	Desc.+Traj.
35	GeoCoTDrive [330]	arXiv'26	📷💬🔶📈	NS NV	VLM ViT, DA3/VGGT	LLaVA-1.5/Qwen2.5-VL	LH	Desc.+Traj.
• Sec. 4.1.2: Numerical Action Generator								
36	LMDrive [49]	CVPR'24	📷💬🗺️🔶	CA	ResNet [121]	LLaMA [284] Vicuna [285]	REG	Ctrl.
37	BEVDriver [331]	IROS'25	📷💬🔶	CA	InterFuser [332]	LLaMA-3.1 [292]	REG	Ctrl.+Traj.
38	CoVLA-Agent [43]	WACV'25	📷💬📈	CV	CLIP [95]	LLaMA2 [284]	REG	Desc.+Traj.
39	ORION [54]	ICCV'25	📷💬🔶	NS B2	EVA-02 [307]	Vicuna-1.5 [285]	GEN	Desc.+Traj.
40	SimLingo [61]	CVPR'25	📷💬🔶	CA B2	InternViT [74]	Qwen2 [72]	REG	Ctrl.+Traj.
41	DriveGPT4-V2 [97]	CVPR'25	📷💬🔶🗺️📈	CA	CLIP [95] SigLIP [302]	Qwen [303] Tinyllama [333]	REG	Ctrl.+Traj.
42	AutoVLA [53]	NeurIPS'25	📷💬🔶📈	NS B2 NV WA	Qwen2.5-VL [92]	Qwen2.5-VL [92]	LH	Traj.
43	DriveMoE [334]	arXiv'25	📷💬📈	B2	PaliGemma [335]	PaliGemma [335]	GEN	Ctrl.
44	DSDrive [83]	arXiv'25	📷💬🔶	CA	ResNet [121]	LLaMA [284]	REG	Desc.+Traj.
45	OccVLA [87]	arXiv'25	📷💬📈	NS	VQ-VAE [152]	PaliGemma-2 [336]	REG	Traj.
46	VDRive [89]	arXiv'25	📷💬🔶📈	NS B2	Qwen2.5-VL [92], CVQ [337]	InternVL3 [93]	GEN	Desc.+Traj.
47	ReflectDrive [338]	arXiv'25	📷💬🔶📈	NV	LLaDA-V [326]	LLaDA-V [326]	GEN	Traj.
48	E3AD [339]	arXiv'25	📷💬🔶📈📋	TC	Qwen2.5-VL [92]	Qwen2.5-VL [92]	REG	Traj.
49	LCDrive [340]	arXiv'25	📷💬📈	PA	DINOv2 [169]	Qwen3 [323]	LH	Traj.
50	Alpamayo-R1 [341]	arXiv'25	📷💬🔶📈📋	PR	Cosmos-Reason1 [342]	Cosmos-Reason1 [342]	REG	Desc.+Ctrl.+Traj.
51	UniUGP [343]	arXiv'25	📷💬🔶📈	WA NS NP	Qwen2.5-VL [92]	Qwen2.5-VL [92]	GEN	Desc.+Traj.
52	MindDrive [344]	arXiv'25	📷💬🔶📈	NV	ResNet-34 [121]	LLaVA-1B [73]	GEN	Traj.
53	AdaThinkDrive [345]	arXiv'25	📷💬🔶📈	NV	InternVL3 [93]	InternVL3 [93]	REG	Desc.+Traj.+Meta.
54	Percept-WAM [346]	arXiv'25	📷🌐💬🔶	NS NV	InternViT [74]	InternVL2 [317]	REG	Traj.
55	Reasoning-VLA [347]	arXiv'25	📷💬📈	NS NV WA A2	Qwen2.5-VL [92]	Qwen2.5-VL [92]	REG	Traj.
56	SpaceDrive [348]	arXiv'25	📷💬📈🗺️	NS B2	Qwen2.5-VL [92]	Qwen2.5-VL [92]	REG	Desc.+Traj.
57	OpenDriveVLA [71]	AAAI'26	📷💬🔶📈	NS	BEVFormer [6]	Qwen2.5 [349]	LH	Ctrl.+Traj.
• Sec. 4.2.1: Dual-System: Explicit Action Guidance								
58	DriveVLM [62]	CoRL'24	📷💬📈	NS SU	-	QwenVL [303]	REG	Desc.+Traj.
59	LeapAD [350]	NeurIPS'24	📷💬🗺️🚦	CA	QwenVL [303]	GPT-4 [290], Qwen1.5 [351]	LH	Ctrl.+Traj.
60	FasionAD [99]	arXiv'24	📷💬🔶📈	NS CA	GenAD [103]	CLIP [95], QwenVL [303]	GEN	Ctrl.+Traj.
61	Senna [108]	arXiv'24	📷💬🔶	NS	VADv2 [352], ViT [305]	Vicuna [285]	REG	Desc.+Traj.
62	DME-Driver [98]	AAAI'25	📷💬🗺️🔶📈	HB	UniAD [32]	LLaVA [91]	REG	Desc.+Traj.
63	SOLVE [353]	CVPR'25	📷💬📈	NS	EVA-02 [307]	LLaVA-1.5 [321]	REG	Ctrl.+Traj.
64	ReAL-AD [354]	ICCV'25	📷💬🔶	NS B2	UniAD [32], VAD [33]	MiniCPM-2.5 [355]	REG	Ctrl.+Traj.
65	LeapVAD [356]	TNNLS'25	📷💬🗺️📈	DL CA	QwenVL [303] InternVL2 [317]	GPT-4o [357]	LH	Ctrl.+Traj.
66	DiffVLA [63]	arXiv'25	📷💬🔶📈	NV	CLIP [95]	Vicuna-1.5 [285]	GEN	Traj.
67	FasionAD++ [358]	arXiv'25	📷💬🗺️🔶📈	NS CA	BEVFormer [6]	Vicuna-1.5 [285], QwenVL [303]	GEN	Ctrl.+Traj.
68	LADA [359]	arXiv'26	📷💬🔶📈	B2	InternViT-300M	Qwen2-0.5B	REG	Traj.
• Sec. 4.2.2: Dual-System: Implicit Representations Transfer								
69	VLP [21]	CVPR'24	📷💬🗺️🔶📈	NS	UniAD [32], VAD [33]	CLIP [95]	REG	Traj.
70	VLM-AD [47]	CoRL'25	📷💬🗺️📈	NS	UniAD [32], VAD [33]	CLIP [95]	REG	Traj.
71	DiMA [360]	CVPR'25	📷💬📈	NS	UniAD [32], VAD [33]	LLaVA-1.5 [321]	REG	Ctrl.+Traj.
72	ALN-P3 [361]	arXiv'25	📷💬🗺️	NS	VAD [33]	LLaMA-2 [284]	REG	Desc.+Traj.
73	VERDI [362]	arXiv'25	📷💬🗺️📈	NS	VAD [33]	Qwen2.5-VL [92]	REG	Ctrl.+Traj.
74	VLM-E2E [104]	arXiv'25	📷💬🗺️	NS	EfficientNet [129]	CLIP [95]	REG	Traj.
75	ReCogDrive [45]	arXiv'25	📷💬📈	NV	InternViT [74]	Qwen2.5 [349]	GEN	Desc.+Traj.
76	InsightDrive [64]	arXiv'25	📷💬🗺️📈	NS	ResNet [121]	BERT [363]	REG	Traj.
77	NetRoller [364]	arXiv'25	📷💬	NS DL	CLIP [95]	LLaMA-2 [284]	REG	Traj.
78	ViLaD [365]	arXiv'25	📷💬📋	NS	SigLIP-2 [325]	LLaDA-V [326]	GEN	Ctrl.+Traj.
79	OmniScene [366]	arXiv'25	📷💬🗺️📈	NS	ResNet [121]	Qwen2.5-VL [92]	REG	Traj.
80	LMAD [367]	arXiv'25	📷💬	NS DL	VAD [33]	LLaVA-1.5 [321]	LH	Desc.+Meta.

From an architectural standpoint, current VLA methodologies for autonomous driving can be grouped into:

- **End-to-End VLA**: a single model directly maps multimodal sensory inputs and language to actions.
- **Dual-System VLA**: a VLM provides high-level reasoning or guidance, while a specialized driving module executes fast, low-level action.

4.1 End-to-End VLA for Autonomous Driving

End-to-end VLA frameworks aim to unify perception, reasoning, and planning within a single architecture [368]. By leveraging the generalization ability of multimodal large language models (MLLMs), they directly transform multimodal observations into actions, reducing reliance on hand-crafted modules and task-specific heuristics. According to the form of their outputs, existing approaches can be broadly divided into two families, as illustrated in Figure 5: **textual action generators**, which operate primarily in the language space, and **numerical action generators**, which predict trajectories or controls in a continuous or discretized numeric space.

4.1.1 Textual Action Generator

Textual action generators formulate driving as a reasoning problem in the language space. The model produces human-readable symbolic decisions, allowing it to "think" and justify its outputs in words. Depending on the abstraction level of these outputs, existing methods can be grouped into *meta-actions* and *trajectory waypoints*.

Meta-Actions are discrete, semantic driving decisions, such as "accelerate", "stop", or "change lane". They form an interpretable interface between high-level reasoning in VLMs and downstream controllers. Early works mainly used language models to output free-form text or conceptual descriptions, which are not directly executable. DriveMLM [56] narrows this gap by aligning LLM outputs with behavioral planning states in a modular stack, enabling language models to act as intermediate planners whose symbolic decisions can be converted into control commands.

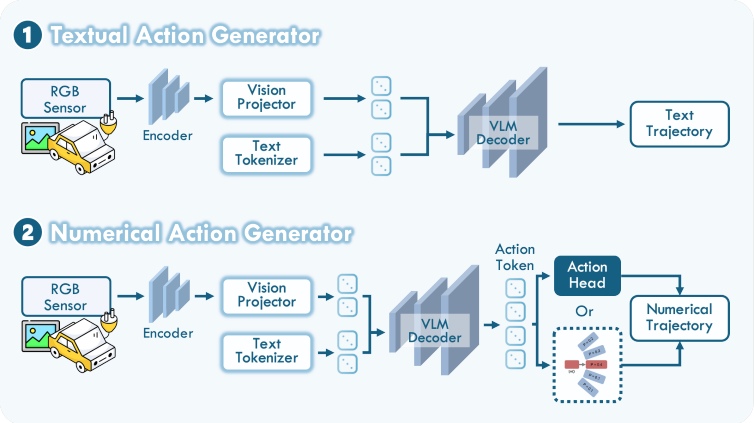


Figure 5 The categorization of **End-to-End VLA models** based on the form of model outputs, including *Textual Action Models* (Sec. 4.1.1), and *Numerical Action Models* (Sec. 4.1.2).

Subsequent methods strengthen robustness and reasoning-planning alignment with reinforcement learning and chain-of-thought supervision [60, 279, 316]. AlphaDrive [279] introduces Group Relative Policy Optimization (GRPO) [60] to refine meta-actions using rewards that jointly consider trajectory quality, decision correctness, and format consistency. DriveAgent-R1 [315] first fine-tunes on a curated CoT dataset to encourage step-wise visual reasoning, then applies RL with trajectory- and meta-action-based rewards to bias reasoning paths toward decisions that are practically useful for driving. Recognizing that single-frame front-view inputs limit temporal and spatial understanding, Sce2DriveX [313] further incorporates multi-view video streams and BEV representations, enabling context-aware meta-decisions that are consistent with road topology and spatiotemporal dynamics. DriveMA [369] rethinks the language interface by compressing decisions into compact one-step meta-actions, while MAGNIFIED [370] extends the GRPO-style recipe with RL fine-tuning of MLLMs that reason before emitting motion plans. To scale this RL paradigm, DriveReward [371] contributes a generative vision-language reward model and dataset for rewarding driving decisions, and OmniDrive-R1 [372] adopts reinforcement-driven interleaved multimodal chain-of-thought for more trustworthy decisions. Several recent systems push the reasoning frontier further: MindDriver [373] introduces progressive multimodal reasoning, C-CoT [374] performs counterfactual chain-of-thought for safer decisions, and an adaptive intend-reflect-refine loop [375] together with a self-reflective Counterfactual VLA [376] iteratively revise their own meta-decisions. EvoDrive [377] casts decision generation as Pareto

evolution among self-improving LLM agents for safety-critical scenes. CritiqueDriveVLM [378] moves from tool-augmented CoT toward verifier-guided reinforcement learning distilled into latent thoughts, mitigating the reasoning hallucinations that standard supervised fine-tuning leaves unchecked, while FactorDrive [379] adapts the depth of multi-step reasoning to planning-critical spatial-physical factors rather than applying uniform reasoning effort to every scene.

Trajectory Waypoints–based textual generators frame motion planning as the prediction of future coordinates expressed in natural language, thereby unifying reasoning and trajectory forecasting within a single linguistic sequence. DriveLM [58] is an early representative of this paradigm, modeling autonomous driving as graph-structured visual question answering and generating textualized trajectory waypoints conditioned on multi-stage perception, prediction, and planning. Building on this idea, subsequent works adopt end-to-end multimodal formulations. EMMA [96] integrates camera observations and navigation commands into a unified language-driven pipeline for joint perception, road-graph understanding, and trajectory prediction. To enhance robustness in challenging scenarios, ImpromptuVLA [280] introduces an 80K-clip corner-case dataset, demonstrating that pretraining on diverse edge cases significantly improves trajectory accuracy and closed-loop stability. LightEMMA [380] further benchmarks 12 vision–language models, revealing clear trade-offs between interpretability and numerical precision. XCoT-VLA [381] pushes this integration further by making the chain-of-thought itself executable, directly binding intermediate reasoning steps to trajectory outputs rather than treating language and trajectory as loosely coupled.

A complementary research direction focuses on better aligning reasoning with decision-making. RDA-Driver [286] enforces consistency between chain-of-thought explanations and trajectory outputs through tailored constraints, while Drive-R1 [316] leverages reinforcement learning to improve alignment between textual reasoning and waypoint prediction. Deferred Exposure [382] tackles verifiability from another angle, withholding future trajectory information during reasoning so that the model's chain-of-thought cannot simply be reverse-engineered from the answer it is meant to justify. Beyond alignment, efficiency and knowledge integration are explored by FastDriveVLA [105] via token pruning, WiseAD [318] through explicit driving priors, and OmniDrive [308] using counterfactual reasoning. MVPruner [383] extends token pruning to the multi-view setting, replacing fixed pruning schedules with dynamic, importance-aware rates. WKER [306] further enhances robustness under occlusion by combining instruction-guided token selection with external knowledge sources. BLUE [384] instead targets efficiency by improving the language use within compact driving VLAs, and a lightweight confidence-aware language model [385] attaches calibrated uncertainty to its textual decisions. Other systems enrich the language interface with explicit critique or risk reasoning: a critic-centric framework [386] judges candidate decisions before driving, HERMES [387] couples risk-aware multimodal reasoning for long-tail scenes, BeLLA [388] grounds an end-to-end BEV language assistant, and a unified formulation [389] jointly handles language-action understanding and generation.

FocusDrive [329] makes the visual referent of reasoning explicit: descriptions of decision-relevant objects are paired with image-patch references before generating action plans and textualized waypoints. This provides a spatial interface for linking a rationale to the image evidence it invokes, complementing methods that constrain reasoning only at the language or trajectory level.

GeoCoTDrive [330] extends spatial grounding by first localizing planning-relevant 2D regions, then retrieving frozen geometric features within those regions. Its nuScenes and NAVSIM variants interleave these features into autoregressive trajectory generation, while the Bench2Drive variant uses a diffusion planner. Its PlanningGrounding dataset supervises the intermediate grounding interface; the current formulation remains limited to front-view observations and depends on grounding accuracy.

To support and stress-test this reasoning-centric paradigm, several resources and analyses have emerged. nuReasoning [390] provides a reasoning-centric dataset and benchmark for long-tail driving, while VLADriveBench [391] probes how chain-of-thought relates to the emitted driving actions. On the analysis side, recent work questions whether the reasoning is genuine—probing the faithfulness of chain-of-causation explanations [392] and exposing vulnerabilities of reasoning-enabled driving VLAs through ReasonBreak [393]. Complementary analyses probe where the reasoning capacity actually resides and what it depends on: depth-wise probing and pruning of the planning token [394] asks how much of a driving VLA's language-model depth the final action truly requires, an interpretability study [395] examines what complex road scenarios look like through the model's own attention, and a data-centric critique [396] argues that today's text-centric VQA and CoT training

data teach VLA models to talk about driving more than to see and act within it. On the deployment side, reasoning-aware speculative decoding [397] accelerates inference without retraining the underlying reasoning process.

Overall, textual action generators offer strong interpretability and rich reasoning but must bridge a fundamental gap between discrete language tokens and continuous control spaces. This mismatch can introduce precision limits and, in extreme cases, unstable or collapsed trajectories.

4.1.2 Numerical Action Generator

Numerical action generators augment VLM backbones with mechanisms that produce directly usable numeric outputs. The model still leverages language-driven reasoning internally, but its final predictions are expressed as trajectories, waypoints, or control values that can be consumed by classical planners or low-level controllers. Two main realizations exist: *additional action heads* attached to the backbone, and *additional action tokens* that discretize continuous actions into a token space.

Additional Action Head. A common strategy is to attach specialized prediction heads to vision-language models [398]. BEVDriver [331] couples a multimodal encoder with a GRU-based head over BEV features, linking language-grounded reasoning with spatial waypoint prediction. CoVLA-Agent [43] uses a lightweight MLP head trained on the CoVLA dataset, demonstrating that joint supervision from trajectories and captions can simultaneously improve interpretability and numeric accuracy. DriveGPT4-V2 [97] augments token-based planning with an MLP that maps multimodal embeddings to continuous trajectories, enhancing sample efficiency while retaining GPT-style reasoning.

To specialize behaviors, DriveMoE [334] employs a Mixture-of-Experts design whose action head dynamically activates experts for skills such as lane following or overtaking. DSDrive [83] proposes a dual-head coordination module, with one head predicting waypoints and another generating reasoning outputs; distillation from larger VLMs keeps the model compact yet interpretable. LMDrive [49] integrates multimodal encoders with an MLP that directly outputs control signals in a closed loop, marking one of the first instruction-following, language-guided end-to-end systems.

Beyond simple MLPs, ORION [54] replaces deterministic heads with a diffusion-based predictor, modeling multi-modal trajectory distributions under uncertainty. SimLingo [61] decouples temporal speed waypoints from geometric path waypoints via a disentangled MLP head, enabling finer-grained control. Masked-VLA-Diffusion [399] pursues a similar generative route with a masked diffusion action head that is both efficient and explainable, while ChainFlow-VLA [400] couples autoregressive reasoning with flow-based causal planning to reach state-of-the-art NAVSIM trajectories. To specialize behaviors further, SAMoE-VLA [401] and VECTOR-Drive [402] extend the Mixture-of-Experts idea with scene-adaptive routing and tightly coupled trajectory experts, AutoMoT [403] introduces an asynchronous Mixture-of-Transformers backbone, and StyleVLA [404] conditions trajectories on driving style. Latent-space designs such as LaST-VLA [405] and LVDrive [406] reason in latent spatio-temporal or visual representations before decoding actions. Questioning the standard 2D-to-3D prediction pipeline itself, PixelPilot [407] identifies this indirection as a source of inherent error and proposes a more direct alternative. S^2-VLA [408] explicitly decouples semantic and spatial processing streams to close the semantic-physical gap between discrete language tokens and continuous control, and WAM-Diff2 [409] distills a hierarchical autoregressive-to-diffusion cascade to make diffusion-based action generation efficient enough for deployment.

A growing line couples action generation with world modeling and unified backbones [410, 411]: VLA-World [412] learns a world model whose generated futures refine the predicted trajectory, while Uni-World VLA [413] interleaves world modeling and planning. Unified VLAs including OneDrive [414], UniDriveVLA [415], and the streaming MindVLA-U1 [416] fold understanding, perception, and action planning into one architecture, the last explicitly outperforming Vision-Action baselines. DriveDreamer-Policy [417] combines a Qwen3-VL backbone with a diffusion action expert, and VLGA [418] adds an explicit geometry stream before flow-based action generation. SpanVLA [419] further improves action bridging by learning from negative-recovery samples. Training and deployment strategies are also actively studied [420]: Drive My Way [421] aligns the policy to personalized preferences, learning from human takeover data [422] post-trains policies on intervention logs, and several works ground VLM reasoning in reinforcement learning for safe, deployable control, including DriveVLM-RL [423] and a modular sim-to-real framework [424].

CLEAR [425] scales closed-loop reinforcement learning for VLA-based E2E driving, and a broader analysis of post-training strategies [426] compares how RL, distillation, and preference-based objectives transfer across both VLA and purely trajectory-generative planners. Finally, analyses interrogate the modality balance of these systems—asking whether visual information truly drives VLA behavior [427]—and the paradigm extends beyond on-road settings to language-guided off-road 3D trajectory planning over traversable terrain [428]. At inference time, two concurrent steering methods bypass retraining altogether: Latent-Centroid Steering [429] applies single-pass classifier-free guidance to close the command-following gap between navigation instructions and predicted trajectories, while attention steering [430] directly reweights the model's visual attention to correct systematic errors without touching its weights.

Recent action-head designs explore different ways to preserve useful pretrained representations. Qwen-Drive-1.0 [431] retains a shared vision-language backbone while attaching a perception head and a planning expert and mixing general multimodal tasks with driving supervision. LaPla [432] projects policy features into a continuous action latent and decodes them with a frozen trajectory decoder, bypassing discrete codebook lookup at inference. RAF-VLA [433] aligns policy hidden states with future-frame representations during training, exploiting future supervision without requiring explicit future-image generation at deployment. FIVE-VLA [434] combines a compact visual interface with recurrent action memory and avoids autoregressive text generation, showing a complementary route toward low-latency numerical planning.

Additional Action Tokens. Instead of explicit heads, some works reuse the language token space to represent actions. AutoVLA [53] discretizes continuous trajectories into a codebook of action tokens, which are autoregressively generated alongside reasoning tokens, thereby unifying semantic reasoning and planning within a single sequence. Reinforcement fine-tuning penalizes redundant reasoning and improves token efficiency. OpenDriveVLA [71] follows a similar token-based paradigm but grounds token generation in a hierarchical alignment between 2D/3D perception and the language model. Structured features are embedded into a unified semantic space, and interaction tokens for the ego vehicle, environment, and other agents are autoregressively decoded into driving actions. AnchorVLA [435] bridges these two families directly, anchoring discrete token-space decisions to continuous trajectory outputs so that long-tail generalization and commonsense reasoning from the discrete side can still ground a precise, executable plan.

Structured action interfaces also support more selective reasoning and persistent context. GRAVA [436] links language references to visual regions and ego states, then emits compact motion primitives with gear and primitive-specific parameters that a deterministic decoder converts into trajectories. CAR-VLA [437] conditions reasoning on both scene complexity and risk, selecting fast intuition, extended deliberation, or a compact hazard-focused reflex response. AD-Memo [438] maintains explicit language-based memory across an episode and trains its memory and driving behavior with distinct credit assignment in semi-closed-loop reinforcement learning. RefineDrive [439] addresses post-training instead: simulator-verifiable failures are converted into corrected trajectory targets, combined with safety-layered reward optimization, while deployment retains direct trajectory prediction.

Numerical action generators are well-suited for downstream control, as their outputs are natively compatible with planners and actuators. However, they typically sacrifice some interpretability and often require substantial supervised data for stable training. When discretized action tokens are used, quantization artifacts can further limit fine-grained control accuracy.

Grounding VLA Planning in Maps, Rules, and Simulated Effects. A cluster of recent work argues that image-only reasoning underdetermines driving decisions and supplies an explicit external referent. Geo-VLA [440] internalizes map semantics so that planning is geometry-aware rather than purely appearance-driven. RedLight-VLA [441] targets a specific and safety-critical failure of behaviour cloning — rare, rule-governed maneuvers at signalized intersections — using expert futures and automatically generated perception targets rather than new annotation. GAPL [442] grounds LLM-produced plans by estimating action effects in simulation before committing, addressing hallucinated reasoning and weak numerical precision directly. A collaborative multi-modality interaction design [443] instead revisits how the modalities are combined, on the observation that most VLA driving models still reduce the task to visual question answering. Multi-agent orchestration over LLM common-sense reasoning [444] explores the complementary direction of composing several reasoners where single-policy control degrades.

4.2 Dual-System VLA

Dual-system VLA frameworks draw inspiration from the dual-process theory popularized by *Thinking, Fast and Slow* [445]. In this paradigm, a VLM serves as the *slow*, deliberative system that performs high-level reasoning, situational assessment, and linguistic inference, while a specialized autonomous driving module acts as the *fast* system responsible for real-time, low-latency trajectory generation and control [446]. By combining these complementary strengths, dual-system frameworks aim to achieve both interpretability and safety-critical reactivity.

Depending on how VLM outputs interact with the specialized planner, existing methods can be categorized into two families: **explicit action guidance** and **implicit representation transfer**, as illustrated in Figure 6.

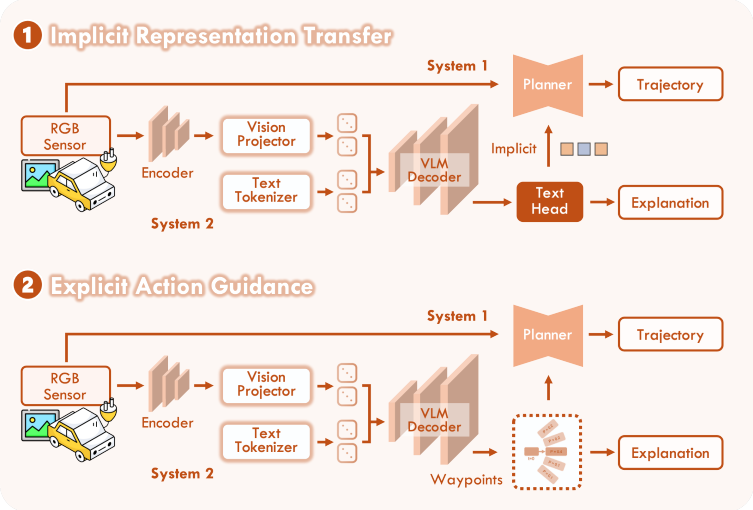


Figure 6 The categorization of **Dual-System VLA Models** based on how VLM interacts with E2E module, including explicit action guidance (Sec. 4.2.1), and implicit representations transfer models (Sec. 4.2.2).

4.2.1 Explicit Action Guidance

Explicit action guidance frameworks use VLMs as structured action generators, whose high-level outputs are subsequently transformed or refined by the fast driving module. These approaches differ in their abstraction level and are mainly grouped into **meta-action guidance** and **waypoint supervision**.

Meta-Action Guidance resorts to VLMs to output symbolic driving intentions, such as "slow down", "change lane", or "turn left", which act as semantic priors for downstream planners. This design leverages the interpretability of linguistic actions while avoiding the precision challenges of directly generating continuous trajectories. Early work such as FasionAD [99] embodies the dual-process design by pairing a fast, data-driven planner with a slow VLM that issues meta-actions; a learned switching mechanism selects the appropriate pathway based on confidence and scene context. LeapVAD [356] refines this structure by combining an analytic branch that builds a memory bank with a heuristic branch that retrieves prior meta-actions for familiar situations.

More recent systems integrate high-level reasoning more tightly with planning. Senna [108] couples a commonsense VLM with an end-to-end planner: Senna-VLM produces natural-language decisions, which Senna-E2E converts into executable trajectories. DiffVLA [63] injects VLM-generated lateral and longitudinal decisions as one-hot priors into a diffusion-based planner, guiding multi-modal trajectory denoising. Senna-2 [447] extends this lineage by explicitly enforcing consistency between the slow VLM decision and the fast end-to-end policy, while DiffVLA++ [448] bridges cognitive reasoning and end-to-end diffusion planning through metric-guided alignment. Decoupled designs such as NaviDriveVLM [449] similarly separate high-level reasoning from motion planning, and OneVL [450] performs one-step latent reasoning and planning while emitting a vision-language explanation.

Hierarchical frameworks such as DME-Driver [98] further separate decision and execution: a VLM-based Decision-Maker supplies meta-decisions or visual attention priors, and a dedicated Executor translates them into fine-grained control. ReAL-AD [354] extends this to a full three-layer hierarchy: strategy, decision, and operation, where VLM-derived situational insights shape progressively refined planning commands.

A related explicit interface can operate through learned intent codes rather than free-form meta-actions. LADA [359] learns a discrete latent intent space from observation–trajectory pairs, trains a language-to-intent translator with limited language annotations, and conditions a separate driving model on the predicted code. Its annotation savings concern the language component; trajectory supervision remains available for policy training. This separation places it within dual-system guidance despite the compact latent interface.

Waypoint Supervision is an explicit guidance that uses VLMs to generate coarse trajectory waypoints, which the fast planning module refines into dense, executable trajectories. DriveVLM [62] adopts a hierarchical reasoning-to-planning pipeline: the VLM produces meta-actions and coarse waypoints through chain-of-thought reasoning, and conventional planners transform them into detailed trajectories.

SOLVE [353] strengthens VLM-planner coordination through a shared vision encoder and a Trajectory Chain-of-Thought module that iteratively refines candidate waypoints before final selection by the E2E planner. These designs provide a tighter numerical interface between reasoning and control, enabling VLMs to influence planning while retaining stability through classical refinement. In a related scoring-based instantiation, SimpleVSF [451] uses a VLM to score and fuse trajectory candidates proposed by a fast E2E planner.

RCVLA [452] extends explicit trajectory guidance to radar-camera sensing. A semantic reasoning stage generates a reference trajectory and structured scene queries, while a second stage refines and selects candidates using retained radar measurements and radar-derived collision-risk estimates. The design distinguishes language-aligned sensor tokens for reasoning from physical measurements used in trajectory arbitration.

A central concern across these systems is the latency gap between the slow VLM and the fast controller, which has motivated several efficiency-oriented architectures. RT-VLA [453] distills a heavy reasoner into a real-time driving model, Fast-dDrive [454] adopts a block-diffusion VLM planner to trade fidelity against speed, and latency-resilient "slow brain, fast planner" designs [455] run the VLM brain alongside a fast real-time planner for urban navigation. In the same spirit, Think at 5 Hz, Act at 20 Hz [456] formalizes this asynchrony as an explicit dual-rate inference schedule, decoupling how often the slow VLM reasons from how often the fast controller must act, and FlashDrive [457] pursues the same latency budget through a dedicated flash-inference implementation of the VLA stack. Analyses of the dual-system premise itself, examining what a VLM reasoner adds beyond a vision-only fast backbone, further clarify this accuracy-cost trade-off [458]. Other works enrich the slow system with world models and structured memory: WorldVLM [459] combines world-model forecasting with vision-language reasoning, CoWorld-VLA [460] reasons within a multi-expert world model, and PRAM-R [461] adds a perception-reasoning-action-memory framework with LLM-guided modality routing. DriveVLA-M0 [462] similarly augments a driving VLA with failure-aware memory, so that past near-misses or mistakes can inform future decisions rather than being forgotten after a single episode. Agentic refinement loops such as CorrectAD [463], a self-correcting agentic system improving E2E planning, and MTRDrive [464], which couples memory and tool use for robust corner-case reasoning, close the loop between reasoning and execution [465]. FIRE-VLA [466] closes a related loop by turning driving failures themselves into training signal for continual self-evolution of the policy.

Overall, explicit guidance approaches maintain strong interpretability and grant VLM a direct role in decision-making. However, they remain sensitive to the accuracy and consistency of VLM outputs; misaligned or ambiguous commands can propagate downstream and degrade planning safety.

4.2.2 Implicit Representations Transfer

Implicit feature constraint refers to methods where the VLM acts as a teacher or auxiliary module during training, transferring reasoning ability or cognitive priors as latent features to the compact E2E network. These approaches fall into two main groups: **knowledge distillation** and **multimodal feature fusion**.

Knowledge Distillation-based approaches transfer VLM-generated explanations, reasoning traces, or structured action semantics into the latent space of the E2E driving model. VLP [21] aligns BEV features and planning queries with pretrained language embeddings using contrastive and supervisory objectives, enabling planners to inherit commonsense scene understanding. VLM-AD [47] generates free-form textual justifications and structured behavior labels using a VLM, distilling them into the planner through an alignment head and an action classification head. This dual-supervision design helps the E2E module acquire richer semantic representations while remaining computationally light during deployment.

More comprehensive alignment is seen in VERDI [362], which aligns perception, prediction, and planning outputs with VLM-generated chain-of-thought explanations, injecting structured reasoning across all stages of the pipeline. ALN-P3 [361] extends this principle with full-stack co-distillation: perception tokens, predicted motions, and planned trajectories are jointly aligned with VLM reasoning to unify cognition and execution. Outcome-Guided Distillation [467] instead lets task outcomes, rather than intermediate reasoning traces

alone, supervise the teacher-student transfer, directly rewarding distilled reasoning that improves downstream driving performance.

Multimodal Feature Fusion-based approaches directly integrate VLM-derived features into the fast planner. InsightDrive [64] introduces language-guided scene representations, where VLM-generated descriptions highlight critical regions and modulate BEV features via cross-attention. VLM-E2E [104] explicitly models driver attention by fusing textual attention cues with BEV features through a learnable gating mechanism. Beyond attention cues, NetRoller [364] extracts latent reasoning variables from VLMs and adapts them into compact features suitable for real-time planners. ReCogDrive [45] aligns linguistic priors with a diffusion-based planner, refining trajectories through reinforcement learning to promote safety and human-like behavior. ETA [140] focuses on efficiency: VLM reasoning is computed asynchronously in earlier frames and fused into current features using an action-mask mechanism, ensuring guidance without incurring high real-time costs.

Implicit transfer methods reduce inference cost and avoid dependence on large VLMs at runtime, but they may sacrifice interpretability, and excessive distillation can oversimplify reasoning signals. Their effectiveness also depends strongly on how well the distilled or fused features align with the capacity of the fast driving module.

5 Datasets & Benchmark

Standardized datasets and benchmarks form the empirical foundation of VLA research, supporting model development, training, and evaluation. Since VLA driving systems integrate perception, language, and action, VLA datasets exhibit substantial diversity in modality composition, annotation granularity, and task definitions. Accordingly, evaluation protocols vary substantially, encompassing conventional trajectory-based metrics, language-centric assessments, and interactive closed-loop evaluations.

5.1 Datasets for VLA in Autonomous Driving

Traditionally, VA datasets provide rich sensory observations (cameras, LiDAR, RADAR) paired with control actions, enabling end-to-end mapping from images to trajectories [76, 118, 468]. These datasets underpin the development of early IL/RL-based VA models.

As language becomes an increasingly important modality for reasoning, instruction following, and explainability, VLA datasets have emerged [56, 58, 62, 473]. These datasets extend traditional driving logs with textual instructions, question-answer pairs, or rationales aligned with visual observations and expert actions [43, 280]. In general, a dataset is considered VLA-compatible when it provides temporally or semantically aligned language annotations that connect visual observations with actions or trajectories, enabling tri-modal learning. The summarized collections are provided in Table 4.

5.1.1 Vision-Action Datasets

Originally, BDD100K [468] provides 100K diverse driving videos from across the United States, covering a wide spectrum of weather, lighting, and traffic conditions, making it a foundational dataset for behavioral cloning and end-to-end driving. Later, nuScenes [76] offers 1,000 multi-sensor driving scenes with synchronized 6-camera surround views, LiDAR sweeps, radar, 3D boxes, and motion trajectories, supporting both perception tasks and multi-agent motion forecasting. Larger-scale datasets such as the Waymo Open Dataset [278] and Argoverse 2 [78] further extend this paradigm with higher-resolution sensors, longer trajectories, and detailed HD maps, enabling robust training of perception-to-prediction pipelines in diverse urban settings. Complementing these efforts, nuPlan [118] incorporates long-horizon ego trajectories, dense map context, and simulation interfaces for closed-loop testing, providing comprehensive supervision for evaluating decision-making and planning under complex, real-world conditions. While lacking explicit language supervision, these VA datasets establish the visual–action foundation for VLA development by providing structured supervision that links visual perception, temporal dynamics, and expert decision-making.

5.1.2 Vision-Language-Action Datasets

Building upon the visual–action foundation established by VA datasets, VLA datasets enrich driving logs with structured or free-form natural language to support joint perception-language-action learning.

Table 4 Summary of existing **Datasets & Benchmarks** for training and evaluating the VA and VLA models.
- **Vision Sensor Inputs**: Camera, LiDAR point cloud, RADAR point cloud, and Map.
- **Vision Types**: Real: Data collected from real driving scenes, and Sim: Data collected from simulator.
- **Language Annotation Types**: A: Automatic labeling process, and M: Manual labeling process.
- **Action Types**: Traj.: Numerical trajectory output, and Ctrl: Control signal output.
- **Action Metrics**: Open: Open-loop Evaluation, CL: Closed-loop Evaluation, and Lang.: Language-based Evaluation.

Scales denote frames unless units are given; annotation elements are not interchangeable with QA pairs. doPlan reports unique driving hours. DrivingBench evaluates model–harness systems on one low-speed private course with context-sharing retries.

Dataset	Year	Vision			Language			Action		Other Tasks
		Sensor	Type	Scale	Category	Label	Scale	Type	Metric	
• Vision-Action Datasets										
BDD100K [468]	2020		Real	120M	-	-	-	Traj.	-	Percept.
nuScenes [76]	2020		Real	1.4M	-	-	-	Traj.	Open	Percept.
Waymo[77]	2020		Real	200M	-	-	-	Traj.	Open	Percept., Forecast.
nuPlan [118]	2021		Real	4.6M	-	-	-	Ctrl, Traj.	CL	Forecast.
Argoverse 2 [78]	2021		Real	300K	-	-	-	Traj.	Open	Percept., Forecast.
Bench2Drive [114]	2024		Sim	2M	-	-	-	Traj.	CL	-
RoboBEV [18]	2025		Real	866K	-	-	-	Traj.	Open	Percept.
WOD-E2E [469]	2025		Real	800K	-	-	-	Traj.	Open	-
• Vision-Language-Action Datasets										
BDD-X [275]	2018		Real	8.4M	Caption	M	26K	Ctrl	Open	Reason.
Talk2Car [470]	2022		Real	400K	Caption	M	12K	Ctrl, Traj.	Open	Ground.
SDN [276]	2022		Sim	-	Instruction, QA	A+M	8.4K	Ctrl, Traj.	CL	QA
DriveMLM [56]	2023		Sim	-	Reason., Deci.	A+M	-	Ctrl, Traj.	CL, Lang.	Reason., QA
LMDrive [49]	2024		Sim	3M	Instruction	A+M	64K	Traj.	CL	-
DriveLM-N [58]	2024		Real	4.8K	QA	M	445K	Ctrl, Traj.	Open	Reason., QA
DriveLM-C [58]	2024		Sim	64K	QA	A	3.76M	Ctrl, Traj.	Open	Reason., QA
HBD [98]	2024		Real,Sim	-	Deci., Descrip., QA	A+M	-	Traj.	Open	Descrip., QA
VLAAD [277]	2024		Real	-	Reason., QA	A+M	64K	Ctrl.	Lang.	Caption, QA
SUP-AD [62]	2024		Real	-	Action, Reason., QA	A+M	-	Ctrl, Traj.	Open, Lang.	Reason, QA
NuInstruct [471]	2024		Real	11.8K	Instruction	A	91K	Ctrl	Lang.	Reason.
WOMD-Reason [472]	2024		Real	63K	QA	A	2940K	Plan.	Lang.	Reason., QA
DriveCoT [473]	2024		Sim	-	CoT, Deci.	A	36K	Ctrl	Open	Reason.
Reason2Drive [474]	2024		Real	-	Reason., QA	A	632K	Ctrl, Traj.	Open	Reason., QA
DriveBench [19]	2025		Real	19.2K	QA	A+M	20.5K	Ctrl	Lang.	QA
MetaAD [279]	2025		Real	120K	Reason., Plan, QA	-	30K	Ctrl	Lang.	Reason.
OmniDrive [308]	2025		Real	-	Reason., QA	A	-	Ctrl, Traj.	Open	Reason.
NuInteract [475]	2025		Real	34K	Caption, QA	A	1.5M	Ctrl	Lang.	Percept., Ground.
DriveAction [476]	2025		Real	2.6K	QA	A	16.18K	Ctrl	Lang.	-
ImpromptuVLA [280]	2025		Real,Sim	2M	Instruction, QA	A+M	80K	Ctrl, Traj.	Open, CL	QA
CoVLA [43]	2025		Real	6M	Caption	A	6M	Traj.	Open	-
OmniReason-N [320]	2025		Real	-	QA	A	-	Ctrl, Traj.	Open	Reason., QA
OmniReason-B2D [320]	2025		Sim	-	QA	A	-	Ctrl, Traj.	Open	Reason., QA
CARScenes [477]	2025		Real	5.2K	-	A+M	-	-	-	-
DSBench [478]	2025		-	98K	-	-	-	-	-	-
RoboDriveBench [479]	2025		Sim	64.6K	-	-	-	Traj.	-	-
ScenePilot-4K [480]	2026		Real	27.7M	-	A	-	Traj.	-	-
Bench2Drive-VL [481]	2026		Sim	-	-	A	-	Ctrl	-	-
Vega [482]	2026		-	100K	-	-	-	Traj.	-	-
V2X-QA [483]	2026		Real	-	QA	M	-	-	-	QA
CrashSight [484]	2026		Real	250	QA	-	13K	-	-	QA
LTD [485]	2026		Real	-	QA	A+M	11.6K	-	-	QA
TRIP-Evaluate [486]	2026		-	837	-	-	-	-	-	-
DriveSpatial [487]	2026		-	-	QA	A+M	15.6K	-	-	QA
PedestrianQA [488]	2026		Real	-	-	-	-	Traj.	-	QA
GeoDrive-Bench [489]	2026		-	-	QA	M	5.05K	-	-	QA
ReactSim-Bench [490]	2026		Real,Sim	2.6K	-	-	-	Traj.	-	-
EventDrive [491]	2026		-	-	-	-	-	Traj.	-	-
DriveJudge [492]	2026		-	33.6K	-	M	-	Traj.	-	-
VL-Evidence [493]	2026		Real	73	QA	A+M	122	-	-	QA
doPlan [494]	2026		Real	50.9 h	Instruct.	M	5.2K	Traj.	Open	Ground.
CASCADE [495]	2026		Real	2.1K clips	Reason.	M	34K elem.	-	-	Causal
AnchorReasoning [496]	2026		Real	416.1K	Reason., Ground.	A+M	395.4K elem.	Traj.	Open	Ground.
CausalDriveBench [497]	2026		Real	-	QA	A	7.3K	Traj.	Open	Causal
DriveHierarchy [498]	2026		Real,Sim	84.3K	QA	-	76.8K	Ctrl.	Open,CL	Ground., Reason.
TrafficSignBench [499]	2026		Sim	29K scenes	-	-	-	Traj.	CL	Rule compl.
PlanningGrounding [330]	2026		Real,Sim	-	Ground.	A+M	146K	Traj.	Open,CL	Ground.
• Preliminary Real-Vehicle Evaluation										
DrivingBench [500]	2026		Real	1 course	Instruct.	-	-	Ctrl.	CL	Latency

Table 5 Comparisons of state-of-the-art models for **Open-Loop Planning** on the nuScenes [76] benchmark.

- **Input**: :Camera, :LiDAR, : Prompt, : Instruct., : Scene Descrip., : Status, : Rule, : Context.
- **Action**: LH: Language Head, RL: Policy w/ Reinforcement Learning, REG: Decoder + MLP, SEL: Trajectory Selection w/ Cost, and GEN: Trajectory Generation w/ Generative Model.
- **Evaluation Metrics**: L2 (↓): L2 Error in meters, and CR (↓): Collision Rate.

Model	Year	Input	Vision	Language	Action	L2 ↓				CR ↓			
						1s	2s	3s	Avg.	1s	2s	3s	Avg.
• Vision-Action Models													
ST-P3 [109]	2022		EfficientNet	-	REG	1.33	2.11	2.90	2.11	0.23	0.62	1.27	0.71
UniAD [32]	2022		ResNet	-	REG	0.44	0.67	0.96	0.69	0.04	0.08	0.23	0.12
VAD [33]	2023		ResNet	-	REG	0.17	0.34	0.60	0.37	0.07	0.10	0.24	0.14
OccNet [116]	2023		ResNet	-	SEL	1.29	2.13	2.99	2.14	0.21	0.59	1.37	0.72
BEV-Planner [90]	2024		ResNet	-	REG	0.30	0.52	0.83	0.55	0.10	0.37	1.30	0.59
Drive-WM [149]	2024		ConvNeXt	-	SEL	0.43	0.77	1.20	0.80	0.10	0.21	0.48	0.26
GenAD [103]	2024		ResNet	-	GEN	0.36	0.83	1.55	0.91	0.06	0.23	1.00	0.43
OccWorld [160]	2024		ResNet	-	REG	0.43	1.08	1.99	1.17	0.07	0.38	1.35	0.60
DriveDreamer [146]	2024		SD	-	REG	-	-	-	0.29	-	-	-	0.15
SparseAD [134]	2024		ResNet	-	REG	0.15	0.31	0.56	0.34	0.00	0.04	0.15	0.06
GaussianAD [135]	2024		ResNet	-	REG	0.40	0.64	0.88	0.64	0.09	0.38	0.81	0.42
LAW [171]	2024		Swin-T	-	REG	0.24	0.46	0.76	0.49	0.08	0.10	0.39	0.19
SSR [172]	2024		ResNet	-	REG	0.18	0.36	0.63	0.39	0.01	0.04	0.12	0.06
Drive-OccWorld [166]	2025		ResNet	-	REG	0.25	0.44	0.72	0.47	0.03	0.08	0.22	0.11
DriveTransformer [34]	2025		ResNet	-	REG	0.19	0.34	0.66	0.40	0.03	0.10	0.21	0.11
DiffusionDrive [35]	2025		ResNet	-	GEN	0.27	0.54	0.90	0.57	0.03	0.05	0.16	0.08
World4Drive [170]	2025		ResNet	-	REG	0.23	0.47	0.81	0.50	0.02	0.12	0.33	0.16
Epona [156]	2025		DC-AE	-	REG	0.61	1.17	1.98	1.25	0.01	0.22	0.85	0.36
SeerDrive [102]	2025		ResNet	-	SEL	0.20	0.39	0.69	0.43	0.00	0.05	0.14	0.06
GuideFlow [139]	2025		ResNet	-	GEN	-	-	-	-	0.00	0.02	0.18	0.07
• Vision-Language-Action Models													
Agent-Driver [501]	2023		-	GPT-3.5	LH	0.16	0.34	0.61	0.37	0.02	0.07	0.18	0.09
DriveVLM [62]	2024		ViT	QwenVL	GEN	0.18	0.34	0.68	0.40	0.10	0.22	0.45	0.27
DriveVLM-Dual [62]	2024		ViT	QwenVL	REG	0.15	0.29	0.48	0.31	0.05	0.08	0.17	0.10
RAG-Driver [39]	2024		CLIP	Vicuna-1.5	LH	0.34	0.37	0.69	0.40	0.01	0.05	0.26	0.10
Senna [108]	2024		ViT	Vicuna-1.5	REG	0.37	0.54	0.86	0.59	0.09	0.12	0.33	0.18
Doe-1 [293]	2024		Lumina-mGPT	BPE Tokenizer	LH	0.37	0.67	1.07	0.70	0.02	0.14	0.47	0.21
VLP [21]	2024		UniAD, VAD	CLIP	REG	0.30	0.53	0.84	0.55	0.01	0.07	0.38	0.15
VLM-AD [47]	2024		UniAD, VAD	CLIP-ViT	REG	0.24	0.46	0.75	0.48	0.12	0.17	0.41	0.23
OpenDriveVLA [71]	2025		ResNet	Qwen2.5-VL	LH	0.14	0.30	0.55	0.33	0.02	0.07	0.22	0.10
OmniDrive [308]	2025		EVA-02	LLaMA2	LH	0.40	0.80	1.32	0.84	0.04	0.46	2.32	0.94
ORION [54]	2025		EVA-02	Vicuna-1.5	GEN	0.17	0.31	0.55	0.34	0.05	0.25	0.80	0.37
EMMA [96]	2025		Gemini-VLM	Gemini	LH	0.14	0.29	0.54	0.32	-	-	-	-
WKER [306]	2025		EVA-02	LLaMA3	LH	0.14	0.30	0.55	0.33	0.07	0.14	0.32	0.18
Drive-R1 [316]	2025		InternVL2	InternVL2	LH	0.14	0.28	0.50	0.31	0.02	0.06	0.19	0.09
ReAL-AD [354]	2025		ResNet	MiniCPM	REG	0.30	0.48	0.67	0.48	0.07	0.10	0.28	0.15
ReAL-AD [354]	2025		ResNet	QwenVL	REG	0.35	0.53	0.71	0.53	0.09	0.12	0.31	0.17
DiMA [360]	2025		VAD	LLaVA-1.5	REG	0.18	0.50	1.03	0.57	0.00	0.05	0.16	0.08
FasionAD [99]	2025		BEVFormer	VLM+Thinking	GEN	0.19	0.62	1.25	0.69	0.02	0.09	0.44	0.18
InsightDrive [64]	2025		ResNet	VLMs	REG	0.23	0.41	0.68	0.44	0.09	0.10	0.27	0.15
S4-Driver [309]	2025		ViT-G	PaLI3	LH	0.13	0.28	0.51	0.31	-	-	-	-
SOLVE [353]	2025		EVA-02	LLaVA-1.5	REG	0.13	0.25	0.47	0.28	0.00	0.16	0.43	0.20
VERDI [362]	2025		VAD	Qwen2.5-VL	REG	0.36	0.62	0.96	0.65	-	-	-	-
OmniReason [320]	2025		EVA-02	LLaVA-1.5	LH	0.15	0.31	0.57	0.34	0.04	0.18	0.98	0.40
FutureSightDrive [304]	2025		ViT	Qwen2-VL	LH	0.14	0.25	0.46	0.28	0.03	0.06	0.21	0.10
Occ-LLM [86]	2025		-	LLaMA2	LH	0.12	0.24	0.49	0.28	-	-	-	-
FastDriveVLA [105]	2025		Qwen2.5-VL	Qwen2.5-VL	LH	0.14	0.29	0.54	0.33	0.00	0.18	0.70	0.29
AutoDrive-R² [319]	2025		Qwen2.5-VL	Qwen2.5-VL	LH	0.13	0.19	0.25	0.19	-	-	-	-
VDrive [89]	2025		Qwen2.5-VL, CVQ	InternVL3	GEN	0.12	0.26	0.50	0.29	0.03	0.16	0.36	0.18
OccVLA [87]	2025		VQ-VAE	PaliGemma-2	REG	0.18	0.26	0.40	0.28	-	-	-	-
FasionAD++ [99]	2024		GenAD	CLIP, QwenVL	GEN	0.13	0.26	0.45	0.28	0.05	0.08	0.15	0.09
ALN-P3 [361]	2025		VAD	CLIP, LLaMA-2	REG	-	-	-	-	0.05	0.09	0.35	0.16
VLM-E2E [104]	2025		EfficientNet	CLIP	REG	0.28	0.50	0.80	0.53	0.01	0.06	0.20	0.09
NetRoller[364]	2025		CLIP	LLaMA-2	REG	0.38	0.66	1.01	0.68	0.06	0.13	0.30	0.16
OmniScene [366]	2025		ResNet	Qwen2.5-VL	REG	0.28	0.53	0.91	0.57	0.00	0.04	0.19	0.08
Sce2DriveX [313]	2025		OpenCLIP	Vicuna-1.5	LH	0.15	0.33	0.59	0.36	-	-	-	-
dVLM-AD [324]	2025		SigLIP2	LLaDA-V	LH	0.15	0.40	0.68	0.41	-	-	-	-
Percept-WAM [346]	2025		InternViT	InternVL2	REG	0.16	0.33	0.60	0.36	-	-	-	-
Reasoning-VLA [347]	2025		Qwen2.5-VL	Qwen2.5-VL	REG	0.05	0.19	0.41	0.22	0.02	0.06	0.13	0.07
CorrectAD [463]	2025		UniAD, VAD	LLM Agent	REG	0.34	0.60	0.94	0.62	0.05	0.14	0.40	0.20
MindDriver [373]	2026		Qwen2.5-VL, VQ-VAE	Qwen2.5-VL	REG	0.27	0.51	0.82	0.53	0.06	0.12	0.32	0.17
AutoMoT [403]	2026		BAGEL	BAGEL	GEN	0.14	0.29	0.54	0.32	0.01	0.06	0.15	0.07
OneDrive [414]	2026		InternVL3, EVA-02	InternVL3	REG	0.13	0.25	0.46	0.28	0.00	0.12	0.43	0.18
UniDriveVLA [415]	2026		InternVL3	InternVL3	GEN	0.27	0.49	0.77	0.51	0.03	0.10	0.21	0.11
VLA-World [412]	2026		Qwen2-VL	Qwen2-VL	GEN	0.11	0.27	0.52	0.30	0.00	0.03	0.26	0.10
DVGT2 [258]	2026		DINO, Geo. Transformer	-	GEN	0.25	0.67	1.43	0.78	0.00	0.07	0.50	0.19
VLGA [418]	2026		VLM, Geo. Expert	VLM	GEN	0.34	0.83	1.52	0.90	0.00	0.15	0.50	0.22
Masked-VLA-Diffusion [399]	2026		LLaDA-V	LLaDA	SEL	0.70	1.31	2.34	1.28	-	-	-	-
Does-VLA [427]	2026		-	-	-	-	-	-	0.311	-	-	-	-
SaMoE-VLA [401]	2026		InternVL, BEVFormer	InternVL	REG	0.24	0.27	0.35	0.29	0.01	0.18	0.60	0.26
LAST-VLA [405]	2026		VLM, VGGT	VLM, Cosmos	RL	0.17	0.33	0.64	0.38	0.00	0.11	0.42	0.18
FastD-Drive [454]	2026		SigLIP2	LLaDA-V	GEN	0.12	0.33	0.50	0.32	-	-	-	-
GraphWorld [256]	2026		World Model	-	REG	0.29	0.55	0.89	0.57	0.00	0.04	0.20	0.08
GEM [236]	2026		ResNet-50, 4D Gaussian Occ.	-	REG	0.19	0.26	0.35	0.27	0.09	0.14	0.26	0.16
ExploreVLA [260]	2026		Show-o, MAGVIT-v2	Show-o	RL	0.28	0.40	0.65	0.44	0.01	0.05	0.25	0.10
DriveVA [257]	2026		Video Diffusion DiT	-	GEN	0.33	0.76	1.43	0.84	0.00	0.07	0.12	0.06
SparseWorld [247]	2026		ResNet, Sparse Scene Rep.	-	REG	0.32	0.56	0.88	0.59	0.08	0.18	0.46	0.24
Bridging-VLA [253]	2026		ResNet, World Model	-	REG	-	-	0.68	0.42	-	-	0.38	0.16
DriveWorldVLA [259]	2026		InternVL3, Swin-T(BEV)	InternVL3	GEN	0.28	0.58	0.99	0.61	0.00	0.10	0.38	0.16

Representative examples include BDD-X [275], which extends BDD100K with time-aligned human rationales, where annotators describe why the driver performed a specific action. This dataset provides early grounding for language-based explanations. DriveLM [58] constructs graph-structured question-answer pairs based on nuScenes and CARLA scenarios. These QA pairs target conditional reasoning, enabling models to infer high-level intent, spatial relations, and driving decisions. Impromptu VLA [280] aggregates data from eight public driving datasets and supplements them with captions, instructions, and QA pairs aligned with expert trajectories. The focus is on corner cases and long-tailed events. Other datasets, such as LingoQA [502] and CoVLA [43], collect real-world driving videos paired with natural language QA or behavior descriptions, emphasizing spatiotemporal reasoning and human-understandable driving motivations.

More recent efforts broaden both the sensing modalities and the linguistic scope of VLA data. EventDrive [491] pairs event-camera streams with language annotations to support vision-language driving intelligence under high-dynamic-range and motion-blur conditions, while large-scale first-person collections such as ScenePilot-4K [480] and the semantic CARScenes dataset [477] provide dense scene-level descriptions geared toward safe driving. At the transportation-system level, open-ended vision-language corpora have also been assembled to train unified mobility foundation models [485], extending VLA supervision beyond the ego-vehicle perspective. On the instruction-following side, Vega [482] and the doScenes instructed-driving challenge with its DVDrive approach [503] ground natural-language commands directly in driving behavior, supplying instruction-conditioned trajectories for closed-loop control.

Notably, QA-style annotations have emerged as a dominant paradigm for extending driving datasets, serving as a common foundation for training and evaluating reasoning and planning capabilities [58, 476]. This paradigm has since diversified along several axes: cooperative and infrastructure-aware reasoning in V2X-QA [483], spatiotemporal intelligence in DriveSpatial [487], region-specific multimodal reasoning in GeoDrive-Bench [489], and safety-critical understanding through pedestrian intention and trajectory prediction in PedestrianQA [488] and crash-scene reasoning in CrashSight [484]. Other benchmarks probe where multi-view models draw their evidence, assessing view-level visual grounding in driving MLLMs [493]. The paradigm continues to broaden along further axes: incident-centric dashcam understanding in AUTOPILOT VQA [504], adverse-weather multimodal understanding with explicit observability awareness in ObsDriveBench [505], and interpretable collision anticipation through concept-aware risk attention in CARA [506]. Cooperative multi-agent reasoning is likewise gaining a dedicated benchmark and vehicle-to-vehicle VLA baseline in CMU-Drive/V2V-VLA [507], extending the cooperative theme introduced by V2X-QA to a full method-benchmark pairing. Beyond the ego-vehicle's forward-facing cameras, the LAIA dataset [508] pairs closed-loop CARLA driving with synchronized human eye-tracking, opening a new axis for training and evaluating attention-aware, explainable E2E driving models. VLN-AVP [509] extends the vision-language-navigation formulation to a distinct driving sub-task, zero-shot autonomous valet parking in unmapped environments. However, the scope and assumptions embedded in such annotations naturally influence model behavior, motivating further exploration of more diverse perspectives, planning horizons, and evaluation protocols for real-world deployment.

New datasets sharpen the distinction between instruction grounding, explanation quality, and executed behavior. doPlan [494] supplies human instructions over extended driving contexts, supporting temporally localized maneuver following beyond short-horizon waypoint prediction. CASCADE [495] annotates spatiotemporal causal structure in driving clips and evaluates generated reasoning against that structure, while AnchorReasoning [496] grounds chain-of-thought annotations in critical visual elements and their implications for driving actions. The human-annotated structure in CASCADE and the model-assisted visual grounding in AnchorReasoning provide different forms of supervision and should not be treated as interchangeable causal ground truth. PlanningGrounding [330] complements these resources with semi-automatic region annotations for planning-critical cues, including road boundaries, conflict areas, and occluded regions. Its 146K annotations connect visual localization to expert-trajectory supervision rather than establishing causal ground truth.

CausalDriveBench [497] pairs causal question answering with counterfactual trajectory tasks, exposing discrepancies between language-level reasoning and action accuracy in the evaluated models. DriveHierarchy [498] separately assesses visual grounding, contextual memory, reasoning, and closed-loop execution, connecting question-answer evaluation to interactive scenarios. TrafficSignBench [499] adds procedurally generated driving scenarios with executable traffic-sign rule checks. Together, these resources move evaluation from whether a model can describe a scene toward whether it can ground, retain, and execute the relevant constraints.

Cheaper Supervision and Reasoning-Level Annotation. Two new corpora attack the data-cost problem this survey identifies from opposite ends. SkyDrive [510] sidesteps in-vehicle collection entirely, extracting 650K driving samples from 137 hours of drone-based aerial traffic monitoring so that a new city can supervise an agent without a fleet. CAViAR [511] instead deepens annotation rather than widening it, providing human-annotated causal accident video for fine-grained reasoning about *responsibility* — who is at fault and which traffic rule was violated — a target largely absent from existing driving-language benchmarks.

Benchmarks That Score the Planning Output. Recent benchmarks converge on evaluating the driving decision rather than an intermediate representation. DriveDegrade [512] makes this explicit: corruption-robustness suites target detection or BEV perception, not the planning output that actually drives the car, so it measures image-degradation tolerance at the trajectory. V2XBench [513] supplies synchronized ego–roadside sensing with closed-loop evaluation plus a progressively structured cooperative-reasoning VQA set, addressing the absence of closed-loop and language-grounded support in existing V2X benchmarks. SIREN-Bench [514] contributes behaviour-level control over a scenario family that log-replay cannot produce, coupling SUMO traffic logic to CARLA sensing for emergency-vehicle interactions with consistent ground truth.

5.2 Evaluation Metrics

Evaluation metrics differ according to the model's output modality: **trajectory-based** metrics for continuous action prediction and **text-based** metrics for models producing linguistic commands or rationales.

5.2.1 Trajectory-Based Action Evaluation

Trajectory-based outputs are typically evaluated in open-loop and closed-loop settings.

Open-Loop Evaluation. The predicted trajectory is directly compared against expert trajectories without executing in a simulator. Metrics such as L2 error and collision rate [109], along with trajectory-based indicators including Average Displacement Error (ADE), Final Displacement Error (FDE), and Miss Rate (MR) [32], are widely used. These metrics regard human driving demonstrations as the ground truth and formulate planning essentially as an imitation learning task. By measuring the deviation between predicted and expert trajectories, they provide a straightforward way to assess the accuracy of motion prediction.

Closed-Loop Evaluation. Instead, it measures the model's performance when interacting with a simulation environment (*e.g.*, CARLA [110]). Representative metrics include route completion (RC), driving score (DS), and infraction distance (ID). Bench2Drive [114] further considers success rate, efficiency, and comfort. Recent suites extend closed-loop evaluation specifically to VLMs, as in Bench2Drive-VL [481], and assess the reactive behavior of world-model simulators in ReactSim-Bench [490]. Because evaluation itself can be costly and subjective, VLMs are increasingly used as judges of driving quality [492], while LLM-driven scenario rollout with anchor-guided diffusion regeneration synthesizes safety-critical situations for stress-testing planners [515].

DrivingBench [500] provides a preliminary real-vehicle complement: general-purpose VLM agents issue steering and speed commands on a low-speed private cone course, with motion continuing during reasoning. Its results compare model–harness systems and use one independent trial per model with context-sharing retries, limiting statistical and public-road generalization.

Non-Reactive Simulation. NAVSIM [115], built on nuPlan [118], uses a distinct non-reactive simulation protocol: candidate ego trajectories are executed against recorded surrounding-agent motion rather than fully reactive traffic. Its Predictive Driver Model Score (PDMS) aggregates ego progress (EP), time-to-collision (TTC), and comfort (C), with penalties for collisions (NC) and driving inadmissibility (DAC). Such scores characterize behavior under the specified rollout assumptions; they do not by themselves establish deployability in interactive traffic.

5.2.2 Text-Based Action Evaluation

For low-level vehicle control expressed in natural language, evaluation covers both linguistic quality and control effectiveness. Standard text metrics, such as BLEU, ROUGE, and CIDEr, are commonly used to assess the quality of generated language [58, 475], which measures n-gram overlap with human-annotated reference commands. Beyond command accuracy, reasoning quality is assessed through rationale consistency [474] and

Table 6 Comparisons of state-of-the-art models on the WOD-E2E [469] test split.
• **Input**: :Camera, : Sys. Prompt, : Instruct., : Scene Descrip., : Status, : Traffic Rule, : Context Info.
• **Action**: LH: Language Head, RL: Policy with Reinforcement Learning, REG: Decoder + MLP, and GEN: Trajectory Generation w/ Generative Model.
• **Evaluation Metrics**: RFS (Overall/Spotlight) (↓): Rater Feedback Score, ADE 5s/3s (↓): Average Displacement Error.

Model	Year	Input	Vision	Language	Action	RFS(Overall)(↑)	RFS(Spotlight)(↑)	ADE 5s(↓)	ADE 3s(↓)
• Vision-Action Models									
Waymo Baseline	2025		-	-	-	7.53	6.60	3.02	1.32
Swin-Trajectory [469]	2025		SwinT	-	REG	7.54	6.68	2.81	1.21
DiffusionDrive [35]	2025		ResNet	-	GEN	7.69	6.65	2.99	1.31
RAP-DINO [517]	2025		DINO	-	REG	8.04	7.20	2.65	1.17
• Vision-Language-Action Models									
OpenEMMA [300]	2025		Qwen2-VL	Qwen2-VL	LH	5.16	4.71	12.74	6.68
HMVLM [518]	2025		ViT	Qwen2.5-VL	LH	7.74	6.73	3.07	1.33
AutoVLA [53]	2025		Qwen2.5-VL	Qwen2.5-VL	LH	7.56	6.94	2.96	1.35
Poutine [519]	2025		ViT	Qwen2.5-VL	LH	7.99	6.89	2.74	1.21
LightEMMA [380]	2025		Qwen2.5-VL	Qwen2.5-VL	LH	6.52	5.71	3.73	1.71
dVLM-AD [324]	2025		SigLIP2	LLaDA-V	LH	7.63	-	3.02	1.29
DriveMA-4B [369]	2026		Qwen3.5	Qwen3.5-4B	LH	8.079	7.169	2.670	1.166
Hermes [387]	2026		ViT	Qwen-VL-Flash, BGE-M3	REG	6.81	-	1.82	0.82
MindVLA [416]	2026		Qwen3-VL	Qwen3-VL	GEN	7.83	-	2.14	0.92
Fast-dDrive [454]	2026		SigLIP2	LLaDA-V	GEN	7.823	-	2.907	1.254

human preference ratings of language explanations, particularly in datasets following the BDD-X [275] format. To assess driving applications, execution-based metrics are introduced for behavior assessment. SimLingo [61] introduces an action-dreaming benchmark. The corresponding actions are mapped from the input instruction, which is open-loop evaluated using the success rate.

Regardless of output modality, these benchmarks emphasize key aspects of action quality, including accuracy, executability, safety, and intention alignment [516].

5.3 Quantitative Experiments & Analyses

This section reviews quantitative benchmarks for evaluating VLA models across action prediction, planning accuracy, and closed-loop driving performance. Among them, nuScenes [76], NAVSIM [115], and Bench2Drive [114] are the most widely used. More recently, WOD-E2E [469] introduces long-tail, safety-critical scenes with human-preference annotations, enabling more robust assessment of modern E2E and VLA systems. Complementary benchmarks broaden the evaluation spectrum toward robustness, safety, and breadth: RoboDriveVLM [479] stress-tests driving VLMs under corruptions and distribution shifts, safety-readiness suites jointly probe external and in-cabin risks [478], and TRIP-Evaluate [486] extends multimodal evaluation across the wider transportation domain beyond ego-vehicle driving QA. A dedicated lane-perception study benchmarks robustness to environmental illusions [522] such as shadows, reflections, and tire marks that naturally occur in real driving footage yet are largely absent from existing corruption suites, and SSP [523] introduces an event-matched synthetic-to-simulation-to-physical evaluation framework for assessing VLA models' cross-domain transfer.

5.3.1 nuScenes Benchmark

The nuScenes open-loop benchmark evaluates planning quality using trajectory-based metrics, including L2 displacement error and Collision Rate, as summarized in Table 5. Basically, vision-action models such as UniAD [32] reports 0.69m L2 and 0.12 collision rate. Incorporating language generally improves performance by providing semantic cues for safer planning. For instance, Drive-R1 [316] combines supervised CoT alignment with RL finetuning to reach 0.31m L2 and 0.09 collision rate.

Beyond accuracy, recent studies explore the role of language in handling complex and long-tailed driving scenarios. While improvements are most evident in common cases, rare and highly complex situations remain an active area of investigation, motivating the integration of richer reasoning signals and data sources.

From a systems perspective, computational efficiency is an important consideration for practical deployment. Lightweight and efficiency-oriented designs, such as InsightDrive [64] (16.3 FPS) and token-pruned architectures like FastDriveVLA [105], illustrate ongoing efforts to balance model capacity with real-time feasibility. For

cross-domain evaluation, nuScenes highlights generalization to unseen cities and distribution shifts as a key benchmark dimension. Works such as VLP [21] and DiMA [360] examine this setting and motivate complementary strategies including domain adaptation, distillation, and data augmentation.

5.3.2 WOD-E2E Benchmark

The Waymo Open Dataset for End-to-End Driving (WOD-E2E) [469] is a large-scale benchmark designed to evaluate end-to-end driving systems under long-tail, safety-critical scenarios that rarely appear in conventional datasets. It contains 4K segments with high-level routing commands, ego-status signals, and multi-camera views, enabling rigorous assessment of perception-planning coupling. A key contribution of WOD-E2E is the Rater Feedback Score (RFS), which measures trajectory quality based on alignment with human preference annotations rather than logged expert trajectories. As shown in Table 6, RFS (Overall and Spotlight) complements conventional ADE metrics by providing a more human-aligned assessment of driving behavior.

Overall results indicate that while vision–action models achieve stable displacement accuracy, VLA models exhibit more diverse performance. Approaches such as Poutine [519], and dVLM-AD [324] achieve balanced RFS and ADE performance, highlighting the importance of effectively aligning language reasoning with trajectory generation. Figure 7 presents the visualized performance of AutoVLA [53] in the WOD-E2E dataset.

5.3.3 NAVSIM Benchmark

NAVSIM [115], built on OpenScene (a redistribution of nuPlan [118]), evaluates planning through non-reactive simulation with recorded surrounding-agent trajectories. It adopts the PDMS metric, which aggregates multiple driving aspects, including No-Collision (NC), Driving Admissibility (DAC), Time-to-Collision (TTC), Ego Progress (EP), and Comfort (C), offering a holistic assessment of safety, efficiency, and driving smoothness. As shown in Table 7, most methods achieve strong performance on safety-related metrics such as NC and DAC, while TTC and EP serve as more discriminative indicators of planning foresight and long-horizon decision quality. These metrics highlight differences in how models balance safety and progress when interacting with dynamic environments.

Representative vision–action approaches, such as WoTE [101], achieve 88.3 PDMS by integrating a BEV-based world model with reward-guided trajectory selection, demonstrating the effectiveness of structured world modeling under this non-reactive evaluation protocol. Building upon this foundation, language-conditioned methods further enhance decision-making. For instance, AutoVLA [53] improves performance to 99.1 NC and 87.6 EP by leveraging language-driven decision priors and a Best-of-N oracle scoring strategy, illustrating how language supervision can guide trajectory selection and improve long-horizon planning behavior.

Recent analyses further qualify the interpretation of NAVSIM scores. Driving on Memory [524] obtains strong non-reactive results by retrieving prior sensor-rich traversals using global pose without current camera input, but this behavior does not transfer equally well to reactive environments. This is evidence of benchmark-specific location and motion priors, rather than evidence that current perception is generally unnecessary. A metric-validity study [525] also shows that rankings can depend on trajectory execution, subscore mappings, and aggregation. Accordingly, comparisons should specify the benchmark version, split, controller, and rollout protocol. For example, RefineDrive's original-NAVTEST evaluation with v2 extended metrics uses a single-stage setting [439]; its EPDMS should not be pooled with the full two-stage NAVSIM v2 protocol.

5.3.4 Bench2Drive Benchmark

Bench2Drive [114] provides a closed-loop evaluation protocol built on CARLA V2, focusing on success rate and composite driving scores to assess goal-directed driving behavior under interactive settings. Unlike open-loop benchmarks, Bench2Drive explicitly evaluates an agent's ability to execute long-horizon tasks and respond to dynamic environmental feedback.

Recent VLA approaches demonstrate clear benefits from language grounding in this benchmark. For example, SimLingo [61] introduces an action dreaming mechanism that aligns natural language instructions with control sequences, achieving a leading driving score of 85.94, as reported in Table 8. These results indicate

that language-guided reasoning can effectively influence closed-loop decision-making and improve planning performance in interactive driving scenarios.

Taken together with open-loop benchmarks, Bench2Drive highlights the growing importance of language–action alignment in VLA systems, particularly for interpreting high-level goals, guiding long-horizon behavior, and adapting actions under complex, human-centered instructions.

6 Challenges & Future Directions

VLA models mark a shift from modular stacks toward holistic, reasoning-driven driving agents. By leveraging large multimodal backbones, they promise richer environmental understanding, stronger generalization, and more interpretable decision-making. Yet, realizing their full potential in safety-critical autonomous driving requires addressing several fundamental challenges. In parallel, emerging research directions point toward next-generation systems that are more efficient, trustworthy, and capable of long-horizon reasoning.

6.1 Current Challenges

6.1.1 Model Architecture and System Efficiency

Real-Time Processing and Latency. VLA models inherit the substantial computational footprint of modern vision-language backbones. High-resolution, high-frame-rate camera inputs generate long visual-token sequences, and multi-view fusion amplifies memory and latency costs. Meeting the strict real-time constraints of autonomous vehicles, therefore, remains difficult [132, 526, 527]. Recent advances in streaming token compression and adaptive visual encoders [105, 528, 529] offer promising directions, alongside token pruning and sparsification schemes such as Prune2Drive [530] and ETA-VLA [531], which combine token pruning with temporal fusion to accelerate inference. A complementary line of work adaptively shortens computation at runtime through dynamic and navigation-guided early exiting [532–534], while Orion-Lite [535] distills the reasoning of large language models into a lightweight vision-only driving model. An orthogonal strategy examines whether onboard computation is necessary at all: a feasibility study of cloud-offloaded VLA inference [536] quantifies the network and scheduling constraints of sharing GPU capacity for models whose forward pass approaches 60 TFLOPs, at the cost of introducing communication latency and connectivity dependence. FIVE-VLA [434], for example, reports approximately 30 FPS on an A100 but approximately 4 FPS on a T4. Thus, latency claims require the hardware, sensor configuration, batch size, and measured processing stages to be stated explicitly; model throughput alone does not establish an end-to-end deployment latency guarantee.

The placement of foundation-model computation also matters. ViRA [145] transfers visual representations through training-only teachers without changing the deployed VA planner, whereas GeoCoTDrive [330] retrieves geometric features during inference. DrivingBench [500] further exposes observation staleness when the vehicle continues moving during reasoning. These settings call for separate reporting of training cost, deployed model cost, and the full observation-to-action delay.

Lack of Domain-Specific Foundation Models. General-purpose VLMs [74, 92, 93] provide strong priors but are not optimized for driving-specific perception, physics, or multi-sensor fusion. Autonomous driving requires precise spatial reasoning, adherence to traffic rules, and an understanding of rare, high-stakes edge cases – abilities not fully captured by generic models. As highlighted in Section 3, dedicated driving foundation models [537] remain a missing cornerstone for scalable and dependable VLA systems.

Table 7 Comparisons of state-of-the-art models for **Non-Reactive Simulation** on the NAVSIM [115] *navtest* benchmark.
• **Input**: Camera, LiDAR, Prompt, Instruct., Scene Descrip., Status, Rule, Context.
• **Action**: LH: Language Head, RL: Policy with Reinforcement Learning, REG: Decoder + MLP, SEL: Trajectory Selection w/ Cost, and GEN: Trajectory Generation w/ Generative Model.
• **Evaluation Metrics**: NC: No At-Fault Collision, DAC (↑): Drivable Area Compliance, TTC (↑): Time-To-Collision, Comf. (↑): Comfort, EP (↑): Ego Progress, and PDMS (↑): Predictive Driver Model Score.

Model	Year	Input	Vision	Language	Action	NC(↑)	DAC(↑)	TTC(↑)	Comf(↑)	EP(↑)	PDMS(↑)
• Vision-Action Models											
TransFuser [31]	2022		ResNet	-	REG	97.7	92.8	92.8	100	79.2	84.0
UniAD [32]	2023		ResNet	-	REG	97.8	91.9	92.9	100	78.8	83.4
VADv2 [352]	2024		ResNet	-	REG	97.2	89.1	91.6	100	76.0	80.9
PARA-Drive [132]	2024		ResNet	-	REG	97.9	92.4	93.0	99.8	79.3	84.0
LAW [171]	2024		Swin-T	-	REG	96.4	95.4	88.7	99.9	81.7	84.6
DRAMA [14]	2024		ResNet	-	REG	98.0	93.1	94.8	100	80.1	85.5
DiffusionDrive [35]	2024		ResNet	-	GEN	98.2	96.2	94.7	100	82.2	88.1
WoTE [101]	2025		ResNet	-	SEL	98.5	96.8	94.9	99.9	81.9	88.3
World4Drive [170]	2025		ResNet	-	REG	97.4	94.3	92.8	100	79.9	85.1
DrivingGPT [155]	2025		VQ-VAE	-	REG	98.9	90.7	94.9	95.6	79.7	82.4
AD-R1 [168]	2025		-	-	RL	98.7	97.8	94.8	100	87.5	91.9
SeerDrive [102]	2025		VoVNet	-	SEL	98.8	98.6	95.8	100	84.2	90.7
Epona [156]	2025		DC-AE	-	REG	97.9	95.1	93.8	99.9	80.4	86.2
GoalFlow [137]	2025		VoVNet	-	GEN	98.4	98.3	94.6	100	85.0	90.3
TrajDiff [131]	2025		ResNet	-	GEN	98.1	97.0	94.3	100.0	82.7	88.5
DiffusionDriveV2 [142]	2025		ResNet	-	GEN	98.3	97.9	94.8	99.9	87.5	91.2
NaviHydra [143]	2025		ResNet	-	SEL	98.7	98.6	88.7	96.2	100.0	92.7
Mimir [144]	2025		ResNet	-	GEN	98.2	97.5	94.6	100	83.6	89.3
WorldDrive [253]	2026		CogVideoX	-	SEL	98.4	96.2	95.1	100.0	81.9	88.1
DriveVA [257]	2026		Wan2.2 VAE/DiT	T5	GEN	99.2	97.5	98.7	100	83.5	90.9
Metis [252]	2026		Wan2.2 VAE/DiT	T5	GEN	98.4	97.2	97.7	-	87.8	89.5
DVGT-2 [258]	2026		DINOv3 ViT-L	-	GEN	97.8	97.2	93.9	100	83.4	88.6
GraphWorld [256]	2026		ResNet-34	-	REG	99.0	97.1	95.5	100	83.2	90.1
Unified Driving Tokens [248]	2026		DINOv3-B	-	SEL	98.7	98.2	95.9	100.0	87.3	91.8
EponaV2 [231]	2026		DINO-Tok	Qwen3-VL-4B	GEN	98.6	97.9	95.7	100	84.8	90.4
IDOL [250]	2026		ResNet-34	-	SEL	98.8	97.6	95.9	100	83.8	90.0
• Vision-Language-Action Models											
ReCogDrive [45]	2025		InternViT	Qwen2.5-VL	GEN	98.2	97.8	95.2	99.8	83.5	89.6
AutoVLA [53]	2025		Qwen2.5-VL	Qwen2.5-VL	LH	99.1	97.1	97.1	99.9	87.6	92.1
ReflectDrive [338]	2025		LLaDA-V	LLaDA-V	GEN	99.7	99.5	99.1	99.9	88.9	94.7
AdaThinkDrive [345]	2025		InternVL3	InternVL3	REG	99.1	98.8	97.2	100.0	87.9	93.0
Percept-WAM [346]	2025		InternViT	InternVL2	REG	98.8	98.6	94.4	99.5	84.8	90.2
Reasoning-VLA [347]	2025		Qwen2.5-VL	Qwen2.5-VL	REG	97.8	93.2	98.1	99.8	80.7	91.7
DiffVLA [448]	2025		CLIP ViT-L/ VoVNet-99	Vicuna-v1.5	REG	-	-	-	-	-	49.12
SimpleVSF [451]	2025		V2-99/ EVA-L/ ViT-L	Qwen2-VL	SEL	98.21	99.29	98.57	-	81.30	-
MTRDrive [464]	2025		CLIP	Qwen2.5-VL	LH	97.3	95.8	91.2	100.0	86.8	88.3
CorrectAD [463]	2025		VAE	T5	REG	-	-	-	-	-	84.7
DriveWorld-VLA [259]	2026		InternVL3, ResNet-34	InternVL3	REG	99.1	98.2	96.1	100.0	85.9	91.3
HybridDriveVLA [458]	2026		InternViT	InternLM2	SEL	98.6	98.6	96.2	100	87.3	92.1
UniWorld [413]	2026		MagVIT-v2	Phi-1.5	REG	98.7	96.7	96.1	100.0	83.2	89.4
LaST-VLA-8B [405]	2026		InternViT-300M	Qwen2.5-7B	LH	98.7	97.9	95.6	100	86.8	91.3
DynVLA [251]	2026		Emu3	Emu3	LH	98.6	98.7	95.5	100	86.8	91.7
DriveMA-4B [369]	2026		Qwen3.5	Qwen3.5-4B	LH	98.7	97.8	95.5	99.9	86.7	91.2
IRR-Drive-4B [375]	2026		Qwen3-VL-4B	Qwen3-VL-4B	LH	98	98.3	93.7	100	88.5	91.3
OneDrive [414]	2026		InternViT-300M	Qwen2.5-1.5B	REG	98.4	95.2	94.9	100	81.1	86.8
ChainFlow-VLA [400]	2026		InternViT	InternVL (2B)	GEN	98.8	98.6	96.1	100.0	90.8	93.6
SpanVLA [419]	2026		Qwen2.5-VL-3B	Qwen2.5-VL-3B	GEN	99.1	97.1	95.2	100.0	86.3	90.3
Xiaomi OneVL (AR) [450]	2026		Qwen3-VL ViT	Qwen3-VL-4B	LH	-	-	-	-	-	88.84
DriveDreamer-Policy [417]	2026		Qwen3-VL-2B	Qwen3-VL-2B	GEN	98.4	97.1	95.1	100.0	83.5	89.2
ExploreVLA [260]	2026		MAGVIT-v2	Phi-1.5	REG	98.8	98.4	96.5	99.9	83.5	90.4

Table 8 Closed-loop and Open-loop performance comparison of E2E-AD Methods on the **Bench2Drive** benchmark.
• **Input**: Camera, LiDAR, Prompt, Instruct., Scene Descrip., Status, Rule, Context.
• **Action**: LH: Language Head, RL: Policy with Reinforcement Learning, REG: Decoder + MLP, SEL: Trajectory Selection w/ Cost, and GEN: Trajectory Generation w/ Generative Model.
• **Evaluation Metrics**: DS (↑): Driving Score, SR (↑): Success Rate. Avg. L2 (↓): Averaged L2 distance of trajectory.

Method	Year	Input	Vision	Language	Action	Closed-Loop				Open-Loop
						DS↑	SR(%)↑	Efficiency↑	Comfort↑	Avg. L2 ↓
• Vision-Action Models										
TCP [126]	2022		ResNet	-	REG	40.70	15.00	54.26	47.80	1.70
ThinkTwice [520]	2023		ResNet	-	REG	62.44	31.23	69.33	16.22	0.95
DriveAdapter [521]	2023		ResNet	-	REG	64.22	33.08	70.22	16.01	1.01
UniAD-Base [32]	2023		ResNet	-	REG	45.81	16.36	129.21	43.58	0.73
VAD [33]	2023		ResNet	-	REG	42.35	15.00	157.94	46.01	0.91
GenAD [103]	2024		ResNet	-	GEN	44.81	15.90	-	-	-
DriveTransformer [34]	2025		ResNet	-	REG	63.46	35.01	100.64	20.78	0.62
ETA [140]	2025		CLIP	-	REG	69.53	38.64	184.51	28.43	-
WoTE [101]	2025		ResNet	-	SEL	61.71	31.36	-	-	-
GuideFlow [139]	2025		ResNet	-	GEN	75.21	51.36	-	-	-
Raw2Drive [130]	2025		ResNet	-	RL	71.36	50.24	214.17	22.42	-
SparseWorld [247]	2026		ResNet50	-	REG	48.95	18.23	173.21	50.62	0.84
DriveVA [257]	2026		Wan2.2 VAE/DiT	T5	GEN	-	-	-	-	1.33
GraphWorld [256]	2026		ResNet-50	-	REG	51.55	25.47	181.12	56.59	0.79
• Vision-Language-Action Models										
ORION [54]	2025		EVA-02	Vicuna-1.5	GEN	77.74	54.62	151.48	17.38	0.68
AutoVLA [53]	2025		SigLIP	Qwen2.5-VL	LH	78.84	57.73	146.93	39.33	-
SimLingo-Base [61]	2025		InternViT	Qwen2	REG	85.94	66.82	244.18	25.49	-
SimLingo [61]	2025		InternViT	Qwen2	REG	85.07	67.27	259.23	33.67	-
ReAL-AD [354]	2025		ResNet	QwenVL	REG	40.76	10.93	-	-	0.87
ReasonPlan [301]	2025		SigLIP	Qwen	LH	64.01	34.55	180.64	25.63	0.61
DriveMoE [334]	2025		BEV Encoder	LLaMA	REG	74.22	48.64	175.96	15.31	0.31
VDRive[89]	2025		Qwen2.5-VL, CVQ	InternVL3	GEN	66.15	50.51	110.23	22.90	0.55
CoReVLA [328]	2025		Qwen2.5-VL	Qwen2.5-VL	LH	72.18	50.00	145.41	34.35	-
BLUE [384]	2026		InternViT-300M	Qwen2-0.5B	REG	90.58	76.18	-	-	-
CriticVLA [386]	2026		InternViT-300M	Qwen2-0.5B	REG	88.02	73.33	269.24	36.45	-
LinkVLA [389]	2026		InternViT-300M	Qwen2-0.5B	LH	91.01	74.55	255.84	34.62	-
MindDriver [373]	2026		Qwen2.5-VL-3B	Qwen2.5-VL-3B	LH	65.48	39.55	143.21	34.63	-
AutoMoT [403]	2026		Qwen3-VL-4B	Qwen3-VL-4B	REG	87.34	70.00	-	-	-
UniDriveVLA [415]	2026		SigLIP-2	Qwen3	GEN	78.37	51.82	198.86	11.78	0.72
VLGA [418]	2026		SigLIP-2, DVGT-2	Qwen3	GEN	79.08	52.73	194.63	13.06	0.69
VECTOR-Drive [402]	2026		Qwen2.5-VL-3B	Qwen2.5-VL-3B	GEN	88.91	71.82	252.10	34.20	-
LVDrive [406]	2026		EVA-02-L, QT-Former	Vicuna-v1.5	GEN	80.71	58.26	155.77	14.34	0.63
Drive My Way [421]	2026		InternViT-300M	Qwen2-0.5B	RL	82.03	70.95	244.98	28.67	-
LfD [422]	2026		InternViT	Qwen2	REG	89.72	73.73	249.01	30.27	-
DynVLA [251]	2026		Emu3	Emu3	LH	88.34	72.73	-	-	-



<think> This is a complex scenario requiring additional reasoning. Slow Thinking

Scene Description:
The scene is set during the daytime in an urban area with clear weather. The road is undergoing construction, as indicated by orange barriers and traffic cones placed along both sides of the road. The front view shows a straight path ahead with no immediate obstacles. The front left and front right views reveal buildings and parked cars on either side of the road. The back view shows a clear path behind the vehicle with no following traffic. The traffic light ahead is green, allowing vehicles to proceed.

Critical Object Description:
1. Traffic Cones: Located on both sides of the road, these cones mark the boundaries of the construction zone. They are stationary and serve as visual guides for navigating the narrowed lane. **- What they are doing:** Marking the construction area. **- What they will do:** Remain stationary. **- Why they are important:** They help the ego vehicle stay within the designated lane and avoid entering the construction zone.
2. Green Traffic Light: Positioned ahead, it indicates that vehicles can proceed. **- What it is doing:** Allowing vehicles to move forward. **- What it will do:** Stay green unless the traffic signal changes. - Why it is important: It dictates the flow of traffic and confirms that the ego vehicle can continue moving forward.

Reasoning on Intent:
- Firstly, the traffic cones indicate that the ego vehicle must stay within the marked lane to navigate the construction zone safely. This requires the vehicle to maintain a steady course and possibly reduce speed to ensure precise navigation.
- Secondly, the **green traffic light** confirms that the ego can proceed without stopping. However, given the construction and narrowed lane, it is prudent to approach cautiously.
- The ego vehicle has been moving forward at a constant speed in the past 4 seconds, and the current driving command is to go straight. Considering the construction zone and the presence of cones, it would be prudent to decelerate slightly to maintain control and safety.

Best Driving Action:
Move forward with a deceleration. This action allows the ego to navigate the construction zone safely while maintaining control and avoiding any potential hazards posed by the narrowed lane and construction equipment. Decelerating slightly ensures that the vehicle can respond promptly to any unforeseen obstacles or changes in the road conditions. </think>

<answer> The final output action is: <action_119><action_590>... </answer>



<think> This is a complex scenario requiring additional reasoning. Slow Thinking

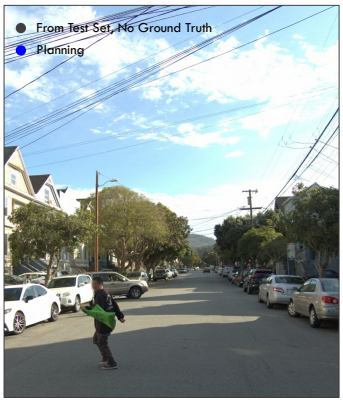
Scene Description:
The scene is a residential area during the daytime with clear weather. The ego vehicle is approaching a T-intersection with a visible "STOP" marking on the road. There are several parked cars along both sides of the street. A few vehicles are visible in the distance, traveling on the intersecting road. No pedestrians are present near the intersection. The traffic appears calm, with no immediate obstacles or hazards.

Critical Object Description:
1. STOP Marking: Located directly in front of the ego vehicle on the road. This indicates that the vehicle must come to a complete stop before proceeding.
2. Vehicles on the Intersecting Road: Positioned further ahead, beyond the intersection.

Reasoning on Intent:
- Firstly, the presence of the "STOP" marking requires the ego vehicle to come to a complete stop. This is a legal obligation and ensures safety by allowing the vehicle to check for any oncoming traffic or pedestrians before proceeding.
- Secondly, observing the vehicles on the intersecting road helps determine when it is safe to proceed. Since they appear to be either stopped or moving slowly, the ego vehicle can likely proceed once it has come to a full stop and confirmed that the way is clear.
- The ego vehicle has been decelerating in the past 4 seconds, indicating an intention to slow down and prepare for the stop. Given the current velocity and acceleration, a quick deceleration is necessary to bring the vehicle to a halt promptly.
- The driving command is to go straight, which aligns with the ego vehicle's current trajectory and the absence of any immediate obstacles or hazards.

Best Driving Action:
Move forward with a quick deceleration. This action ensures that the ego vehicle comes to a complete stop at the "STOP" marking while maintaining control and safety. After stopping, the vehicle can then assess the situation and proceed if it is safe to do so. </think>

<answer> The final output action is: <action_990><action_932>... </answer>



<think> This is a complex scenario requiring additional reasoning. Slow Thinking

Scene Description:
The scene is a residential street during the daytime with clear weather. The road is lined with parked cars on both sides, and there are trees along the sidewalks. Two children are crossing the road in front of the ego vehicle, one on a scooter and the other walking. The front left and front right cameras show parked cars and houses on either side of the street. There are no visible traffic lights or stop signs in the immediate vicinity.

Critical Object Description:
1. Two children crossing the road: The Located in the center of the front view, the child on the scooter is crossing the road while the other child is walking. They appear to be mid-crossing and are likely to finish crossing soon. Their presence is crucial as the ego vehicle must ensure they have safely crossed before proceeding.
2. Parked Cars: Located on both sides of the street, these cars are stationary and do not pose an immediate threat. However, they restrict the lateral movement of the ego vehicle and require caution to avoid any potential collisions.

Reasoning on Intent:
1. Two children crossing the road: The two children are currently crossing the road, and the ego vehicle must wait until they have safely crossed. Once they have finished crossing, the ego vehicle can proceed.
2. Parked Cars: The parked cars limit the lateral movement options. The ego vehicle cannot change lanes or make wide turns without risking collision with the parked cars.
3. Driving Command Instruction: The driving command indicates "go straight", which aligns with the current situation as the ego vehicle is positioned in the center of the road. Given that the two children are expected to finish crossing soon and the road ahead is clear, the ego should prepare to move forward once the children have cleared the path.

Best Driving Action:
Move forward with a quick acceleration. This action is appropriate because once the two children have safely crossed the road, the ego vehicle can quickly accelerate forward to maintain smooth traffic flow while ensuring safety. The quick acceleration helps in efficiently utilizing the road space and maintaining a safe distance from other vehicles. </think>

<answer> The final output action is: <action_0><action_0>... </answer>



<think> This is a complex scenario requiring additional reasoning. Slow Thinking

Scene Description:
The scene is a residential area during a snow-covered day. The road appears to be a two-lane street with houses on both sides. There are no visible vehicles directly in front of the ego vehicle, but there are parked cars along the sides of the road. The front camera shows a clear path ahead with no immediate obstacles. The front left and front right cameras reveal open spaces on either side of the road, with no significant traffic or pedestrians present. The back camera confirms that the road behind is clear as well.

Critical Object Description:
1. Parked Cars: Located on both sides of the road, these cars are stationary and pose no immediate threat to the ego vehicle's movement. However, they should be monitored for any potential opening doors or pedestrians emerging.
2. Residential House: These structures line both sides of the road and do not directly impact the ego vehicle's path. However, they may obscure the view of pedestrians or other vehicles entering the road.

Reasoning on Intent:
1. Clear Path Ahead: The front camera indicates a clear path with no immediate obstacles, suggesting that the ego vehicle can proceed forward without interruption.
2. No Vehicles or Pedestrians: The absence of any vehicles or pedestrians in the immediate vicinity allows the ego vehicle to maintain its current trajectory without needing to adjust speed or direction.
3. **Driving Command Instruction:** The current driving command is "go straight," which aligns with the observed conditions and the vehicle's recent behavior of deceleration followed by stopping.
Given the clear path ahead, lack of nearby traffic, and the driving command to go straight, the ego vehicle should aim to resume a constant speed.

Best Driving Action:
Move forward with a constant speed. This action ensures the vehicle maintains a steady pace while adhering to the driving command and the current road conditions, which are safe and free of immediate hazards. </think>

<answer> The final output action is: <action_337><action_1041>... </answer>

Figure 7 Visualization examples of the AutoVLA [53] reasoning/planning results on WOD-E2E [469] dataset.

6.1.2 Data and Generalization

Generalizing to Rare and Novel Scenarios. One motivation for VLAs is their ability to leverage strong visual-language priors to interpret complex scenes. However, while VLM components may generalize well perceptually, aligning this understanding to the action space introduces new uncertainties. As noted in Section 4, reasoning-rich representations do not automatically translate to robust action generation. Long-tailed scenarios – misbehaving traffic agents, unusual road layouts, unpredictable weather – remain failure points [22, 23, 538–540].

Transfer across camera configurations introduces an additional generalization axis. A zero-shot embodiment study [541] compares data scaling with BEV-Forcing, which grounds visual features in a common geometric frame. The reported benefit of this inductive bias decreases as training camera diversity grows, suggesting that geometric supervision and data diversity should be compared under matched transfer settings rather than assumed to provide independent gains.

Cost of High-Quality Data. VLAs rely on diverse, high-quality multimodal datasets [542, 543], yet collecting paired vision-action-language triplets at scale is expensive. Synthetic environments [544–546] help, but face substantial sim-to-real gaps, with discrepancies in noise characteristics, lighting, and behavior of other agents [8, 18]. Improving data efficiency and mitigating distribution shifts remain long-standing challenges.

6.1.3 Core Capabilities and Trustworthiness

Interpretability & Hallucination. While VLA models produce natural-language rationales via chain-of-thought prompting [53, 54, 547], these explanations are generated artifacts – not faithful reflections of the underlying causal reasoning. Language hallucination [548–550] presents new risks: the model may justify an incorrect decision with a confident but spurious narrative. Such fragility is further exposed under sensor perturbations, where degraded inputs (*e.g.*, fog) silently corrupt the reasoning chain of driving VLAs while leaving the language output superficially plausible [551]. Ensuring consistent grounding between perceptions, actions, and explanations is an open challenge.

The ordering of explanation and action also warrants direct testing. Decide-Then-Explain [552] compares action-first, reasoning-first, and reasoning-free formulations, showing that fluent explanations need not reveal the computational source of a decision. Interventions on reasoning and visual inputs provide more informative evidence than explanation quality alone. Complementarily, a reliability study [553] evaluates visual-corruption effects on driving question-answer accuracy and confidence calibration, finding model- and dataset-dependent gains from visual evidence augmentation. Such QA-level robustness is valuable, but remains distinct from trajectory safety and closed-loop recovery.

Long-Horizon Temporal Coherence. Driving depends on anticipating multi-stage interactions and maintaining situational awareness across extended time horizons [7, 554, 555]. Current transformer-based VLA architectures remain constrained by limited context windows and short-term conditioning, inherited from standard VLM designs. Temporal fragmentation leads to inconsistent decisions, especially in multi-agent or highly dynamic traffic scenes.

Monitoring the Reasoning-Action Link at Runtime. A distinct response to the interpretability problem above is to stop treating the reasoning trace as an explanation and start treating it as a checkable claim. Drive the Thoughts [556] monitors consistency between a VLA's stated reasoning and the trajectory it actually emits, converting the V&V difficulty of black-box planners into a runtime signal. Two complementary lines harden the surrounding system rather than the model: VG-SAF [557] supplies dense heteroscedastic reliability estimates that gate features spatially before an unreliable modality can bias the planner, motivated by asymmetric degradation in which only part of one sensor's field is corrupted; and conformal barrier-function certification [558] certifies the actual safety clearance of the selected trajectory, which candidate-scoring planners do not establish. DriveDegrade [512] makes the same concern measurable, benchmarking image-degradation tolerance on the *planning output* rather than on detection or BEV perception, where corruption-robustness work has concentrated.

6.2 Future Directions

6.2.1 Next-Generation Model Paradigms

Unified Vision-Language-World Models. A promising evolution integrates VLA with predictive world models [94, 171, 559, 560], extending the VA-based models in Section 3.2. Rather than reacting frame by frame, such systems simulate future scene evolution conditioned on candidate actions, enabling proactive planning and more reliable behavior under uncertainty. Building unified, end-to-end world models that jointly reason about perception, language, and dynamics may form the backbone of next-generation autonomous agents.

Richer Multimodal Fusion. As sensor suites diversify, future architectures will incorporate early and tight fusion of LiDAR, Radar, event cameras, and high-definition maps [8, 18, 49]. Language enhances semantic grounding, but robust 3D geometry is indispensable for safe decision-making [87, 119, 561]. Holistic multimodal fusion can combine the interpretability of VLMs with the spatial precision of geometric sensors.

6.2.2 Advancing Intelligence and Adaptation

Socially Aware, Knowledge-Grounded Driving. VLA models must acquire deeper commonsense reasoning – understanding intent, conventions, and causal relationships beyond explicit annotations [306, 562]. Future efforts will draw from large-scale video-language corpora, leveraging external knowledge bases and structured reasoning modules to support socially compliant and anticipatory driving.

Continual & Onboard Learning. Static, offline-trained models cannot capture evolving road infrastructures or regional driving customs [563–565]. Enabling safe, incremental learning from everyday driving, while avoiding catastrophic forgetting and ensuring safety guarantees, is essential for long-term deployment. This relates closely to addressing long-tail generalization gaps.

Table 9 Summary of the evaluation metrics used for evaluating the **trajectory-based** and **text-based** action outputs.

Abbr.	-	Full Name	Description	Ref.
Action-Planning Open-Loop Evaluation				
L2	↓	L2 Error	L2 distance error between the planned trajectory and the human driving trajectory in 3 seconds.	[109]
CR	↓	Collision Rate	How often the self-driving vehicle would collide with other agents on the road.	[109]
ADE	↓	Average Displacement Error	Mean displacement error between predicted trajectories and expert waypoints across the horizon, reflecting overall trajectory accuracy.	[32]
FDE	↓	Final Displacement Error	Displacement error at the final predicted waypoint compared with expert trajectories, emphasizing long-term accuracy.	[32]
MR	↓	Miss Rate	Fraction of prediction time steps where displacement error exceeds horizon-specific thresholds, reflecting failure in trajectory coverage.	[32]
AHE	↓	Average Heading Error	Mean absolute angular deviation between predicted and expert heading over the trajectory horizon, measuring orientation accuracy.	[566]
FHE	↓	Final Heading Error	Absolute angular deviation of predicted heading from expert at the final timestep, reflecting terminal orientation accuracy.	[566]
SLE	↓	Speed L1 Error	Mean absolute error of predicted speed control signals.	[567]
SALE	↓	Steer Angle L1 Error	Mean absolute error of predicted steering angle control signals.	[567]
RFS	↑	Rater Feedback Score	Measure how well the predicted trajectory aligns with human driving preferences by checking whether it falls within trust regions.	[469]
Trajectory-Based Simulation Evaluation				
RC	↑	Route Completion	The percentage of route distance completed.	[175]
DS	↑	Driving Score	RC weighted by a penalty factor that accounts for collisions with pedestrians, vehicles, etc.	[175]
NC	↑	No At-Fault Collision	Penalty for at-fault collisions during the evaluated rollout.	[115]
DAC	↑	Drivable Area Compliance	Boolean evaluation that checks whether the ego vehicle remains inside drivable polygons throughout the rollout.	[115]
TTC	↑	Time To Collision	Boolean verification that the time-to-collision value exceeds safety thresholds, preventing imminent crashes.	[115]
C	↑	Driving Comfort	Compliance with acceleration, jerk, and yaw-rate comfort bounds.	[115]
EP	↑	Ego Progress	Normalized forward progress along the route during the evaluated rollout.	[115]
PDMS	↑	Predictive Driver Model Score	A flexible weighted evaluation score in autonomous driving that aggregates multiple safety, progress, and comfort subscores into a single metric.	[115]
SR	↑	Success Rate	Percentage of navigation episodes that successfully reach the goal within a fixed time budget, indicating overall task completion.	[110]
ID	↑	Infraction Distance	Average driving distance between two infractions, with longer distances reflecting safer and more reliable policy behavior.	[110]
Text-Based Action Evaluation				
CIDEr	↑	Consensus-based Image Description Evaluation	Measures similarity of generated captions to multiple human references using TF-IDF weighted n-grams.	[361]
BLEU	↑	Bilingual Evaluation Understudy	Precision-based metric that compares n-grams of the generated text against reference texts.	[361]
METEOR	↑	Metric for Evaluation of Translation with Explicit Ordering	Considers unigram precision and recall with stemming, synonym matching, and fragmentation penalty.	[361]
Rouge	↑	Recall-Oriented Understudy for Gisting Evaluation	Recall-focused metric using overlapping n-grams, word sequences, or word pairs between generated and reference texts.	[361]
Top-1 Acc	↑	Visual Question Answering Top-1 Accuracy	Percentage of predictions where the most confident output matches the ground truth label.	[361]

6.2.3 Ecosystem for Safe Deployment

Standardized Evaluation & Safety Guarantees. Evaluation metrics from current benchmarks, *e.g.*, NAVSIM [115] and Bench2Drive [114], assess safety and comfort but do not capture key VLA-specific risks such as reasoning failures, instruction-following errors, or cross-modal inconsistencies [548, 549]. Future benchmarks should evaluate multi-step instruction execution, robustness to ambiguous language, and resistance to hallucination; ICR-Drive [568], for instance, probes instruction counterfactual robustness for end-to-end language-driven driving. Beyond empirical testing, formal verification tools are needed to provide theoretical guarantees for safety-critical behaviors. Emerging work moves in this direction by constructing scenario-specific safety envelopes that bound when and how severely a driving VLA may act [569], exposing verifiable meta-action interfaces [369], and supervising decisions at runtime through causal scene narration [570]. Complementary efforts pursue safe alignment via negative-enhanced, risk-aware training [571] and adapt structured assurance arguments through reusable safety-case patterns for VLA-based driving systems [572].

Human-Centric Interaction & Personalization. VLA systems open the door to richer in-car interaction [573, 574]. Natural language enables drivers to specify goals, constraints, and preferences ("*drive cautiously*", "*avoid unprotected left turns*"). Personalization modules [575] can adapt driving styles to different users, enhancing comfort and trust. The challenge lies in balancing personalization with strict safety and regulatory requirements.

7 Conclusion

Vision-Language-Action models are reshaping autonomous driving by coupling perception with high-level reasoning and natural language understanding. This work formalizes the VLA problem setting, outlines the progression from traditional VA pipelines, and organizes existing methods into coherent architectural families together with the datasets and benchmarks that support their development. VLA systems offer clear advantages in interpretability, generalization, and human interaction, but core challenges remain: aligning symbolic reasoning with continuous control, ensuring robustness in long-tail scenarios, and establishing evaluation protocols that faithfully measure instruction following and safety. Progress will depend on advances in efficient architectures, deeper multimodal fusion, world-model-driven planning, and more rigorous human-centered testing. Overall, VLA represents a promising direction for building autonomous agents that are not only competent drivers but also communicative, transparent, and responsive to human intent.

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